Ultrasonic Distance Measurement Circuit Design

Prepared by Minho Eom

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Introduction

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4) Full Schematic

Introduction

1. Project Overview

- **Objective**
  - Design an ultrasonic distance measurement circuit (30–100 cm) with 1 cm resolution.
- **Design Tool:**
  - Simulation & validation in **OrCAD/Pspice**
- **Applications**
  - Automotive **rear parking sensors**, other **proximity detection systems**

2. Measurement Principle & Design Specifications

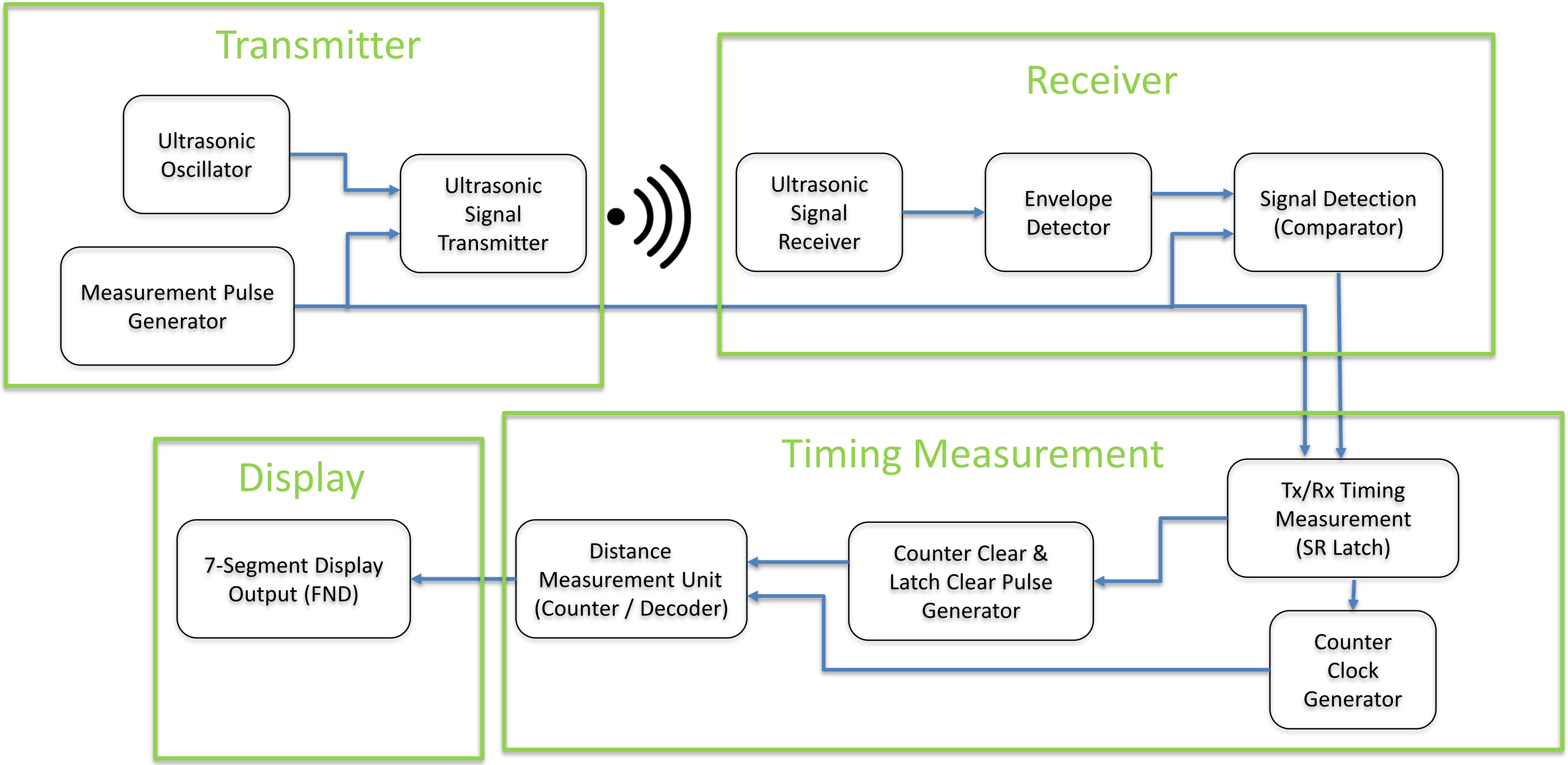
Measurement Principle

- Distance measured from **TOF (time-of-flight)** of ultrasonic waves
- Formula: $\mathrm{d} = \frac{c \times \Delta t}{2}, with\ c\ \approx 340\ m/s\ (at\ room\ temperture)$
- Example: 1 m round trip ≈ **5.8 ms**
- Measure Tx–Rx delay → distance output

Design Specifications

- transmit frequency: **40 kHz ultrasonic burst**
- Burst period: **~65 ms** (~15 Hz)
- Counter clock: **17.2 kHz** (≈ 100 counts per 1 m round trip; 1cm resolution)
- Supply: **12 V DC**
- Display: **3-digit 7-Segment**

3. Block Diagram & Functional Summary (brief)



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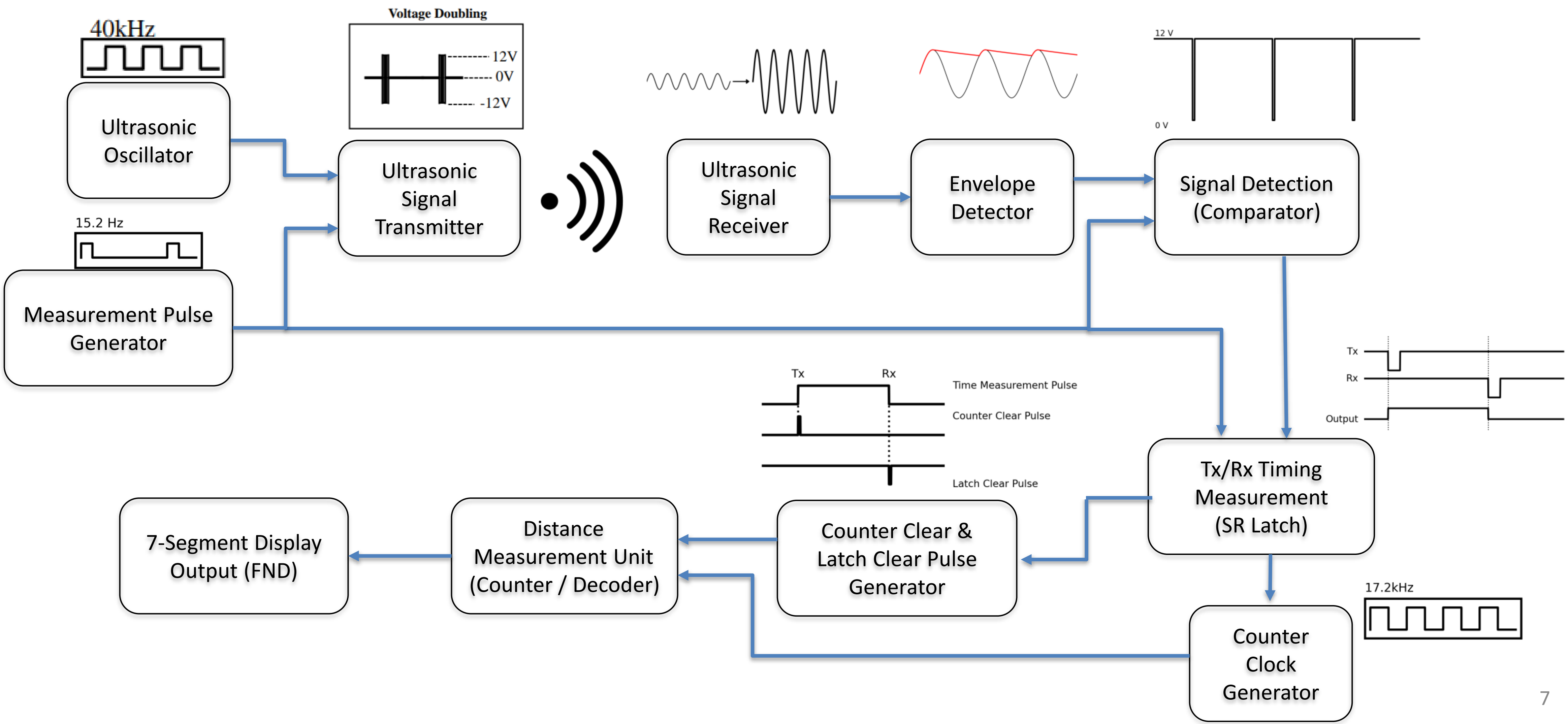
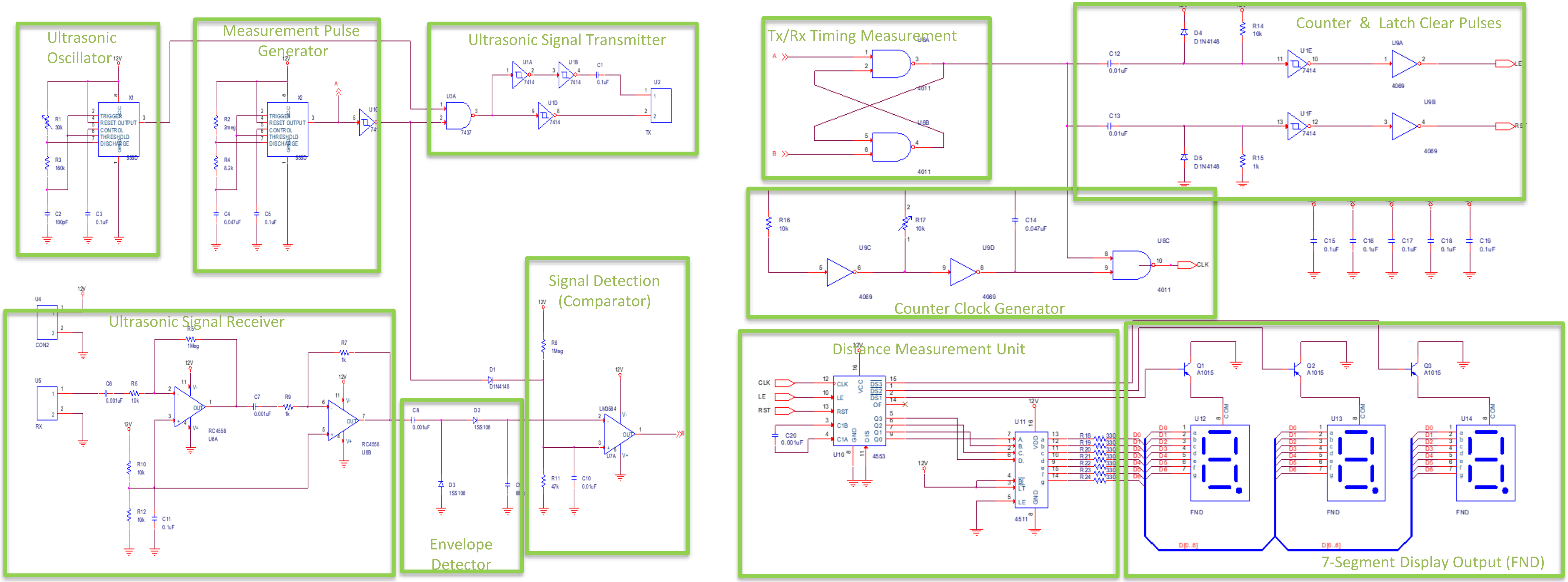


4. Overall Circuit Schematic



5. Bill of Materials (BOM)

No.	Part Name	Spec	Qty
1	Chip IC NE555 (SMD)	–	1
2	IC NE555	–	1
3	IC 4584	–	1
4	IC 4011	–	1
5	IC RC4558	–	1
6	IC LM358	–	1
7	IC 4069	–	1
8	IC 4553	–	1
9	IC 4511	–	1
10	IC Socket 8-Pin	–	3
11	IC Socket 14-Pin	–	3
12	IC Socket 16-Pin	–	2
13	Transistor A1015	–	3
14	FND 500 (7-Segment)	–	3
15	Ultrasonic Sensor TX	SE-400ST160	1
16	Ultrasonic Sensor RX	SE-400SR160	1
17	Diode 1SS106	–	2
18	Diode 1N4148	–	3
19	Connector (2P, Green)	–	1
20	Bare PCB	–	1
21	Resistor	150 kΩ, 1/4 W, 1%	1

No.	Part Name	Spec	Qty
20	Bare PCB	–	1
21	Resistor	150 kΩ, 1/4 W, 1%	1
22	Resistor	10 kΩ, 1/4 W, 1%	8
23	Resistor	2 MΩ, 1/4 W, 1%	1
24	Resistor	8.2 kΩ, 1/4 W, 1%	1
25	Resistor	1 MΩ, 1/4 W, 1%	2
26	Resistor	47 kΩ, 1/4 W, 1%	1
27	Resistor	1 kΩ, 1/4 W, 1%	2
28	Chip Resistor (SMD)	330 Ω (2012)	7
29	Mylar Capacitor	0.047 μF	1
30	Ceramic Capacitor	680 pF	1
31	Ceramic Capacitor	100 pF	1
32	Ceramic Capacitor	0.1 μF	9
33	Ceramic Capacitor	0.01 μF	3
34	Ceramic Capacitor	0.001 μF	4
35	Ceramic Capacitor	0.0047 μF	1
36	Trimmer Resistor	47 kΩ	1
37	Trimmer Resistor	10 kΩ	1

Introduction

1) Ultrasonic Oscillator
2) Measurement Pulse Generator
3) Ultrasonic Signal Transmitter
4) Ultrasonic Signal Receiver
5) Envelope Detector
6) Signal Detection (Comparator)
7) Tx/Rx Timing Measurement (SR Latch)
8) Counter Clock Generator
9) Counter Clear & Latch Clear Pulse Generator
10) Time Measurement Unit (Counter / Decoder) & 7-Segment Display Output (FND)

1. Ultrasonic Oscillator

Block Purpose & Function

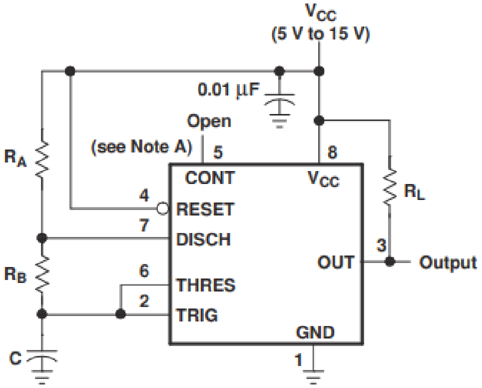
1. Generate a 40 kHz ultrasonic carrier (square wave) to drive the TX stage.

1. Ultrasonic Oscillator

Specification

400ST160	Transmitter
400SR160	Receiver
Center Frequency	40.0±1.0KHz
Bandwidth (-6dB) 400ST160	2.0KHz
Bandwidth (-6dB) 400SR160	2.5KHz
Transmitting Sound Pressure Level at 40.0KHz; 0dB re 0.0002μbar per 10Vrms at 30cm	120dB min.
Receiving Sensitivity at 40.0KHz 0dB = 1 volt/μbar	-61dB min.
Capacitance at 1KHz ±20%	2400 pF
Max. Driving Voltage (cont.)	20Vrms
Total Beam Angle -6dB	55° typical
Operation Temperature	-30 to 70°C
Storage Temperature	-40 to 80°C

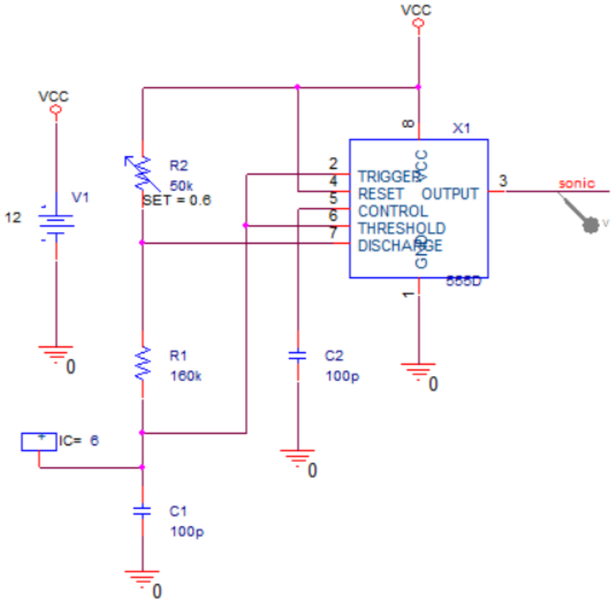
All specification taken typical at 25°C
Closer frequency tolerance can be supplied upon request.



$$t_H \cong 0.693 \times (R_A + R_B) \times C$$

$$t_L \cong 0.693 \times R_B \times C$$

$$f = \frac{1}{T} \cong \frac{1.44}{(R_A + 2R_B) \times C}$$

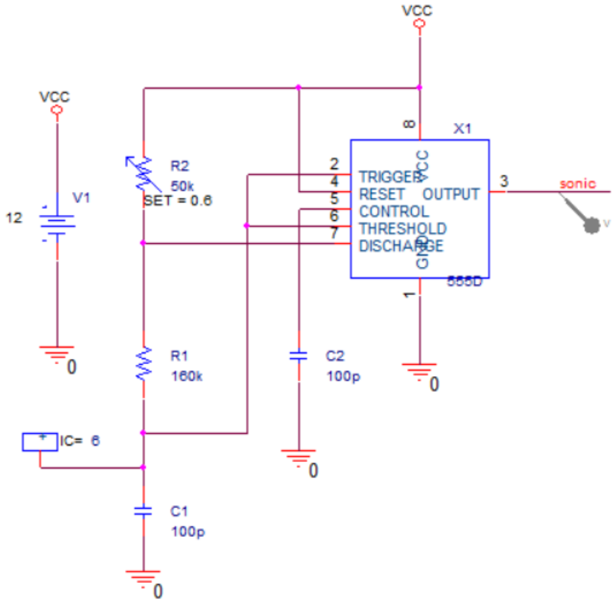


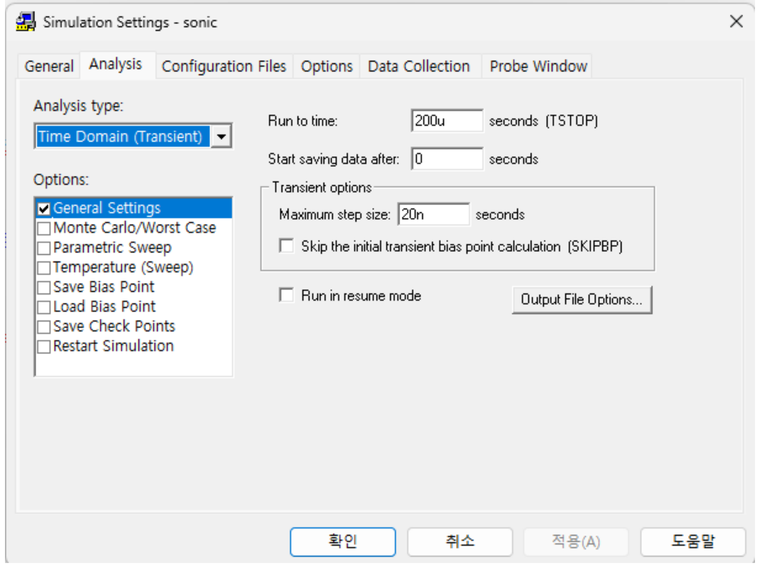
- **Resonant Frequency:** SE-400ST160 requires **40 kHz** for maximum output.
- **Oscillator Design:** Target frequency set to **40 kHz**.
- **NE555 Formula:**

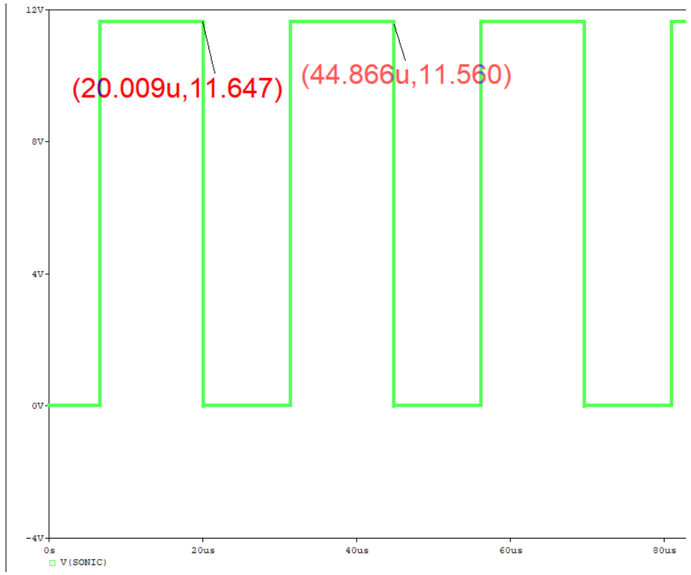
$$F = \frac{1.44}{(R_A + 2R_B) \times c}\ [\text{Hz}],$$

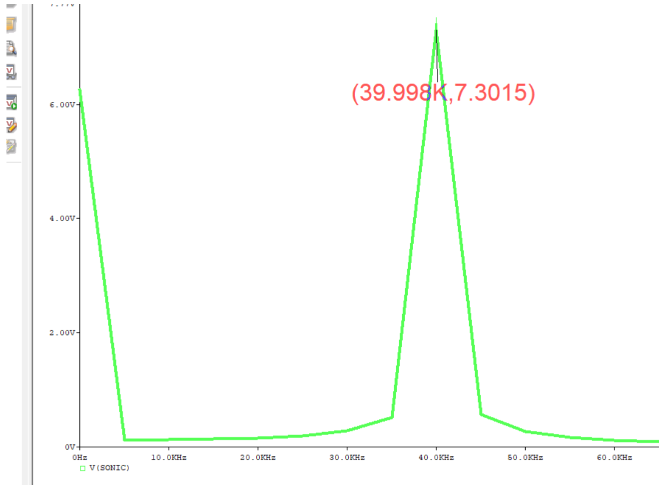
- **Implementation:** Select proper **R1, R2, C1** to achieve 40 kHz.

1. Ultrasonic Oscillator









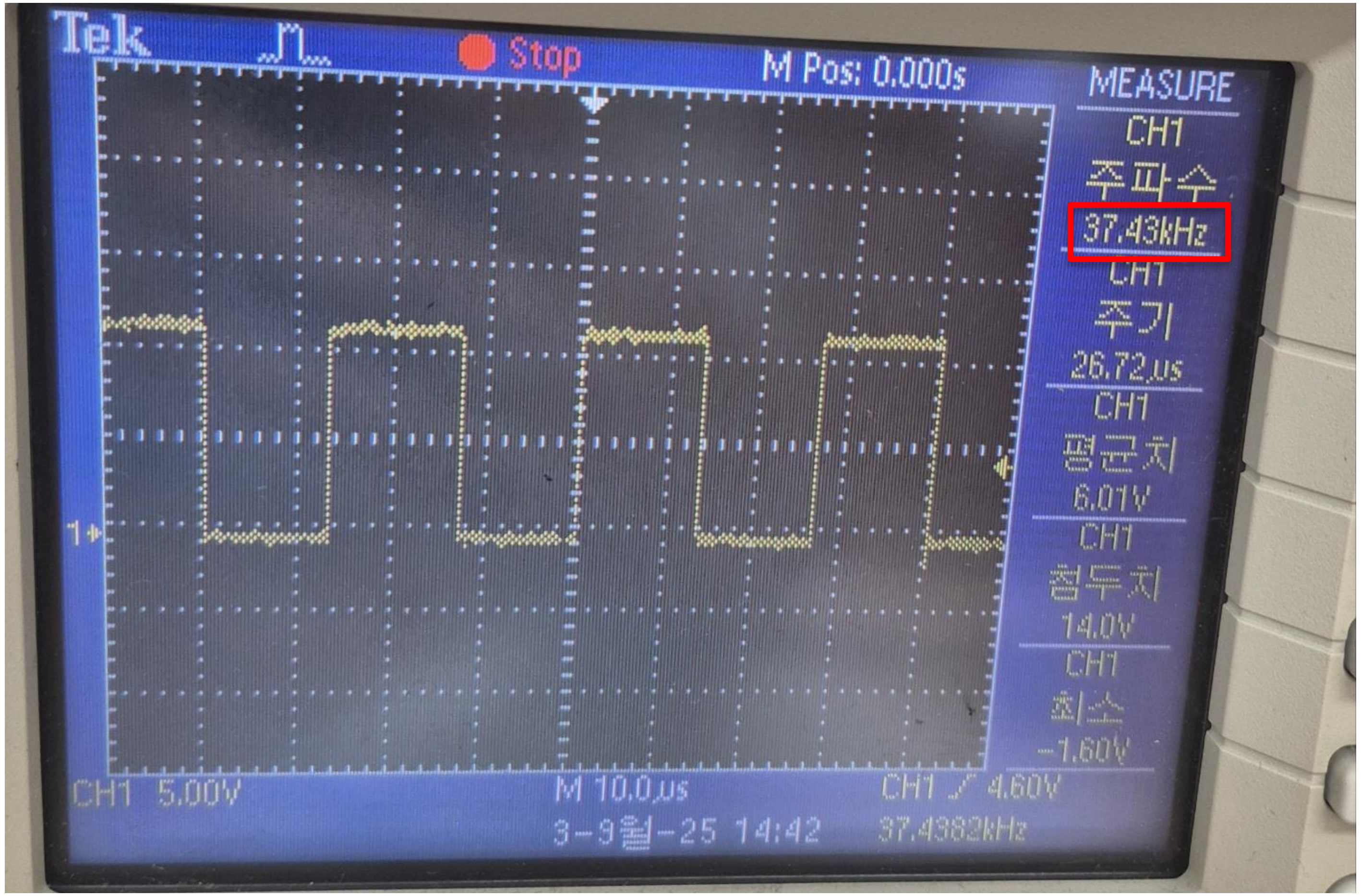
$$(R1 = 30\,k\Omega \qquad R2 = 160\,k\Omega \qquad C1 = 100\,pF)$$

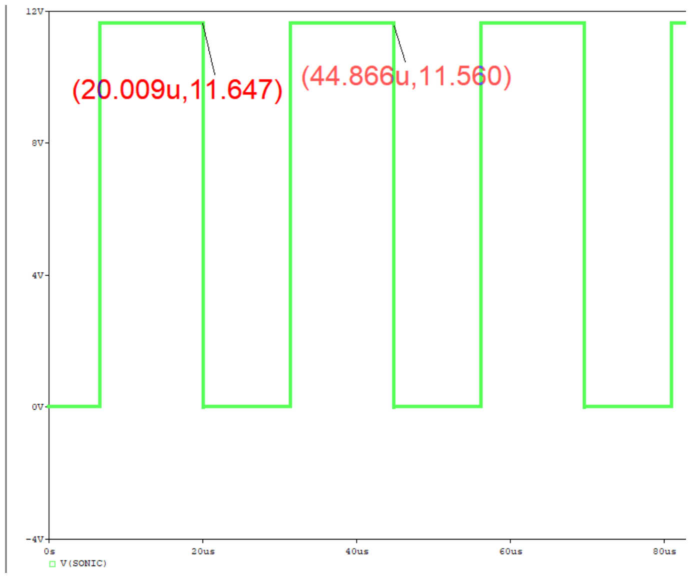
$$t_L = 0.69 \times 160 \times 10^3 \times 100 \times 10^{-12} \approx 11\ \mu sec$$

$$t_H = 0.69 \times (160+30) \times 10^3 \times 100 \times 10^{-12} \approx 13\ \mu sec$$

$$f = \frac{1.44}{\{(30 + 2 \times 160) \times 10^3\} \times 100 \times 10^{-12}} \approx 41.1\ kHz$$

1. Ultrasonic Oscillator



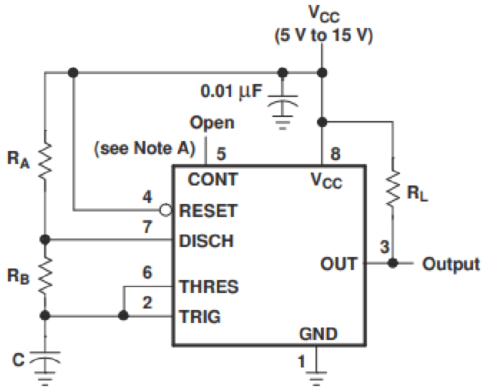


2. Measurement Pulse Generator

Block Purpose & Function

1. Generate timing pulses for ultrasonic burst transmission
2. Produce one pulse approximately every 65 ms (≈15.2 Hz)

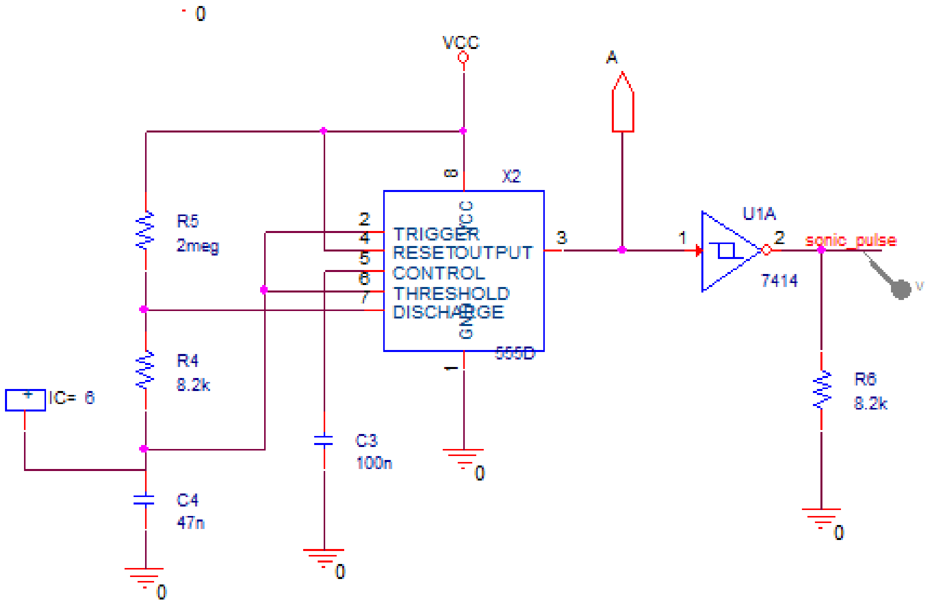
2. Measurement Pulse Generator

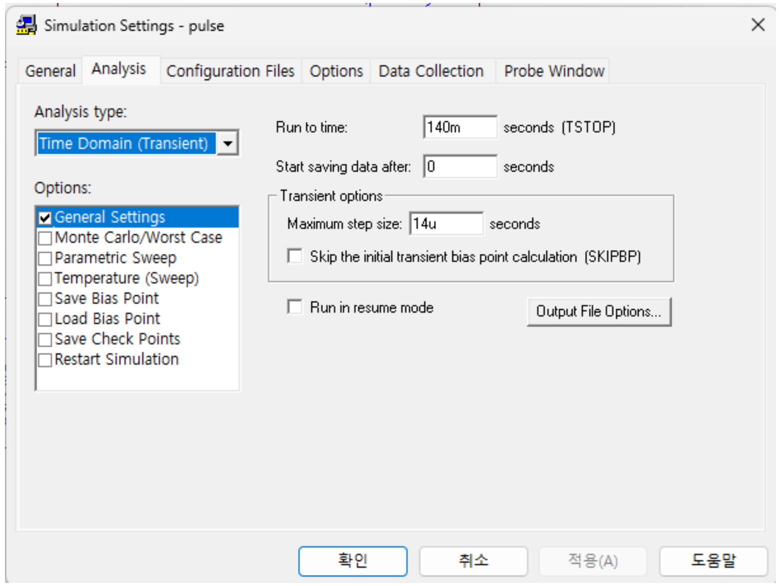


$$t_H \cong 0.693 \times (R_A + R_B) \times C$$

$$t_L \cong 0.693 \times R_B \times C$$

$$f = \frac{1}{T} \cong \frac{1.44}{(R_A + 2R_B) \times C}$$

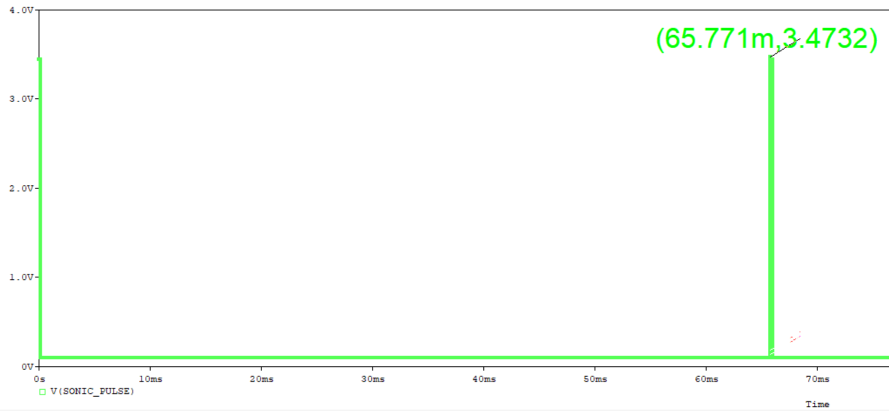




$$t_H = 0.69 \text{ x } (2 \text{ x } 10^6 + 8200) \text{ x } 47 \text{ x } 10^{-9} \approx 65.125\ msec$$

$$t_L = 0.69 \text{ x } 2 \text{ x } 10^6 \text{ x } 47 \text{ x } 10^{-9} \approx 265.92\ usec$$

$$f = \frac{1.44}{\{(2 \times 10^6 + 2 \times 8200\} \times 47 \times 10^{-9}} \approx 15.19\ Hz$$



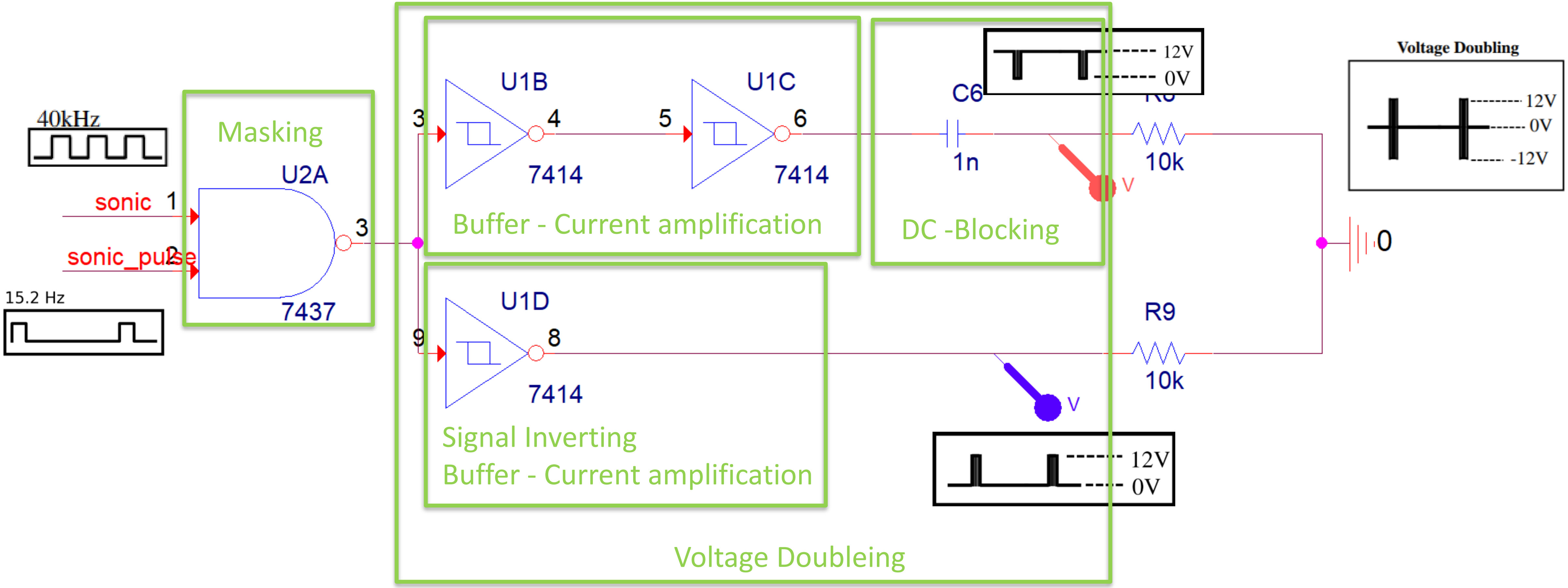
2. Measurement Pulse Generator

3. Ultrasonic Signal Transmitter

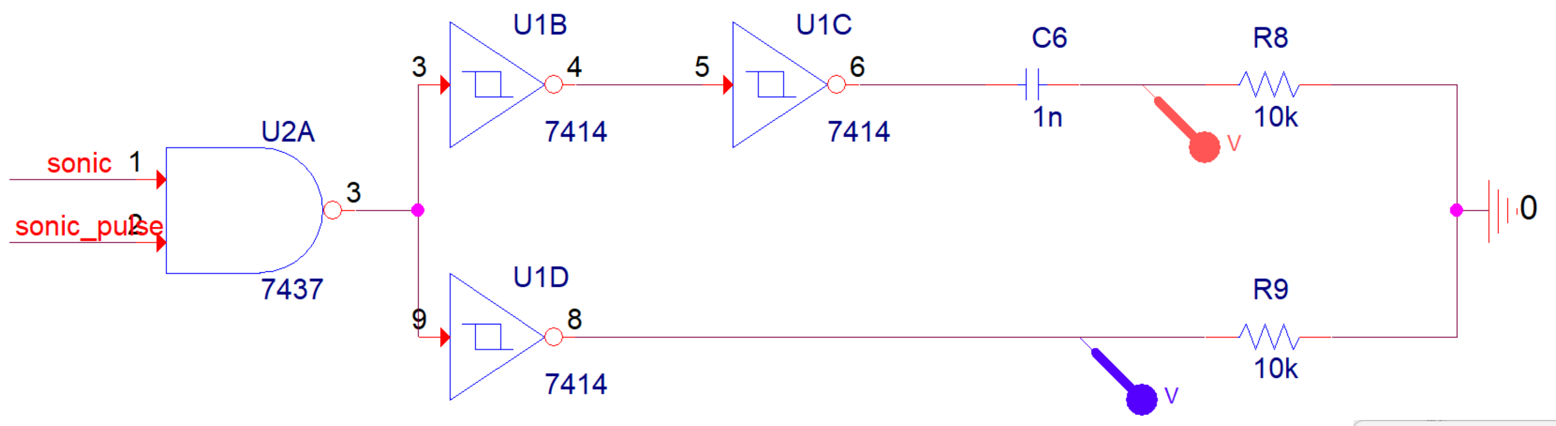
Block Purpose & Function

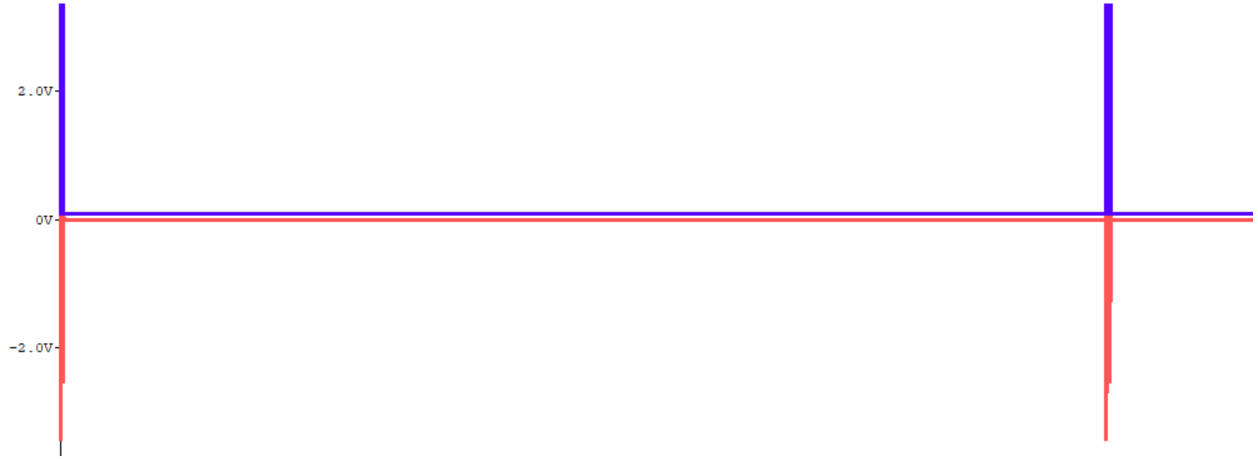
1. Transmit 40 kHz ultrasonic bursts according to timing pulses
2. Use NAND gate masking and buffer/voltage doubler for stronger drive

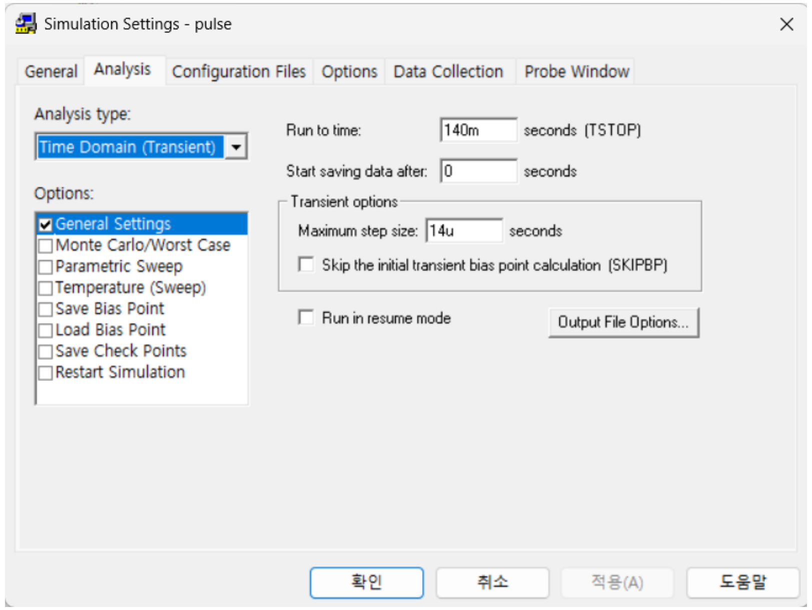
3. Ultrasonic Signal Transmitter



3. Ultrasonic Signal Transmitter







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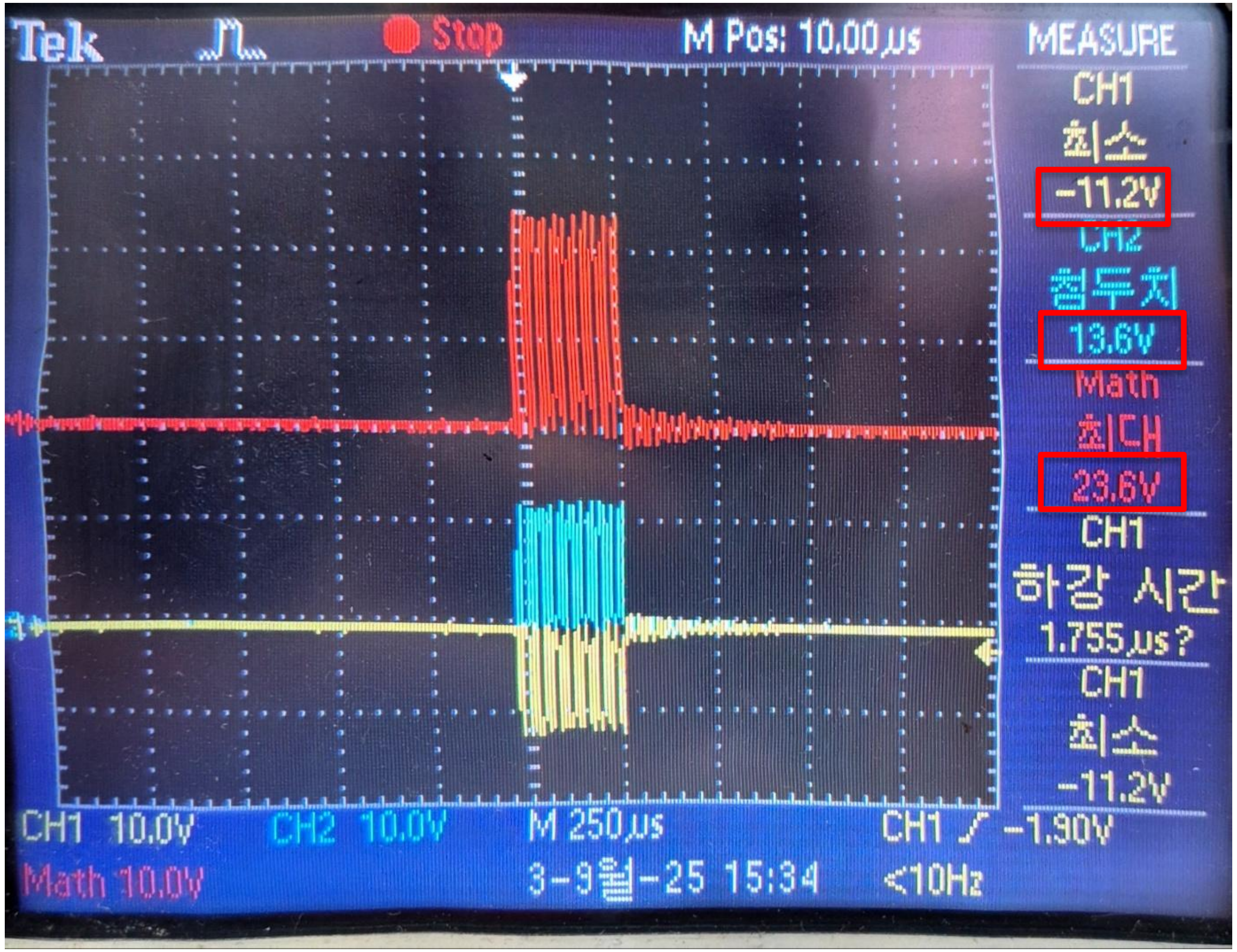
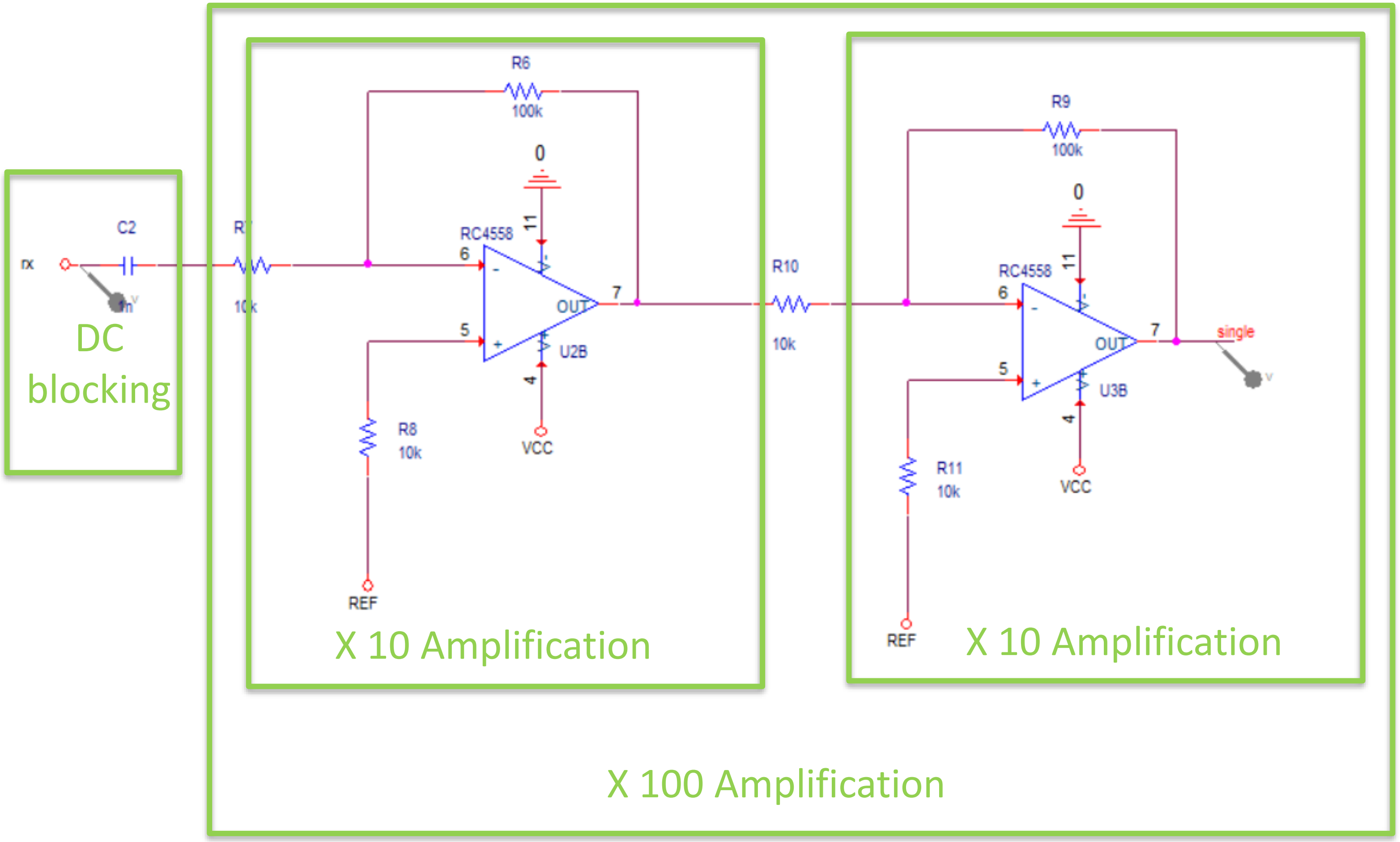


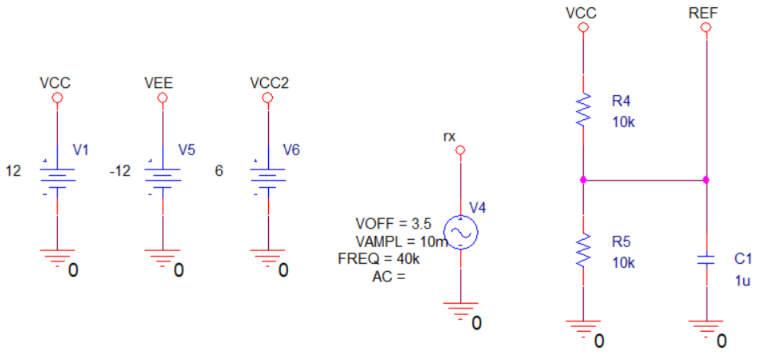
4. Ultrasonic Signal Receiver

Block Purpose & Function

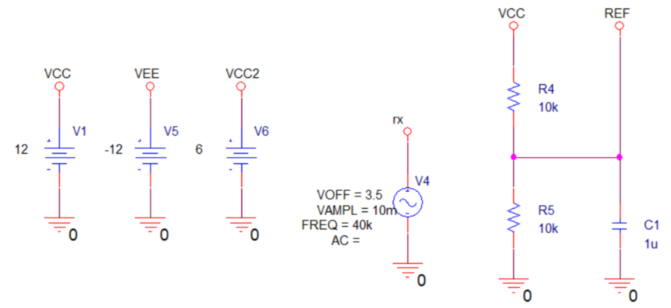
1. Receive reflected 40 kHz ultrasonic signals
2. Amplify weak received signals using a two-stage OP-Amp

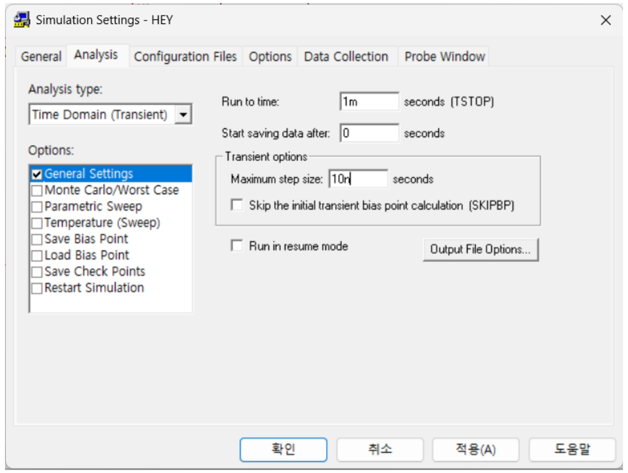
4. Ultrasonic Signal Receiver

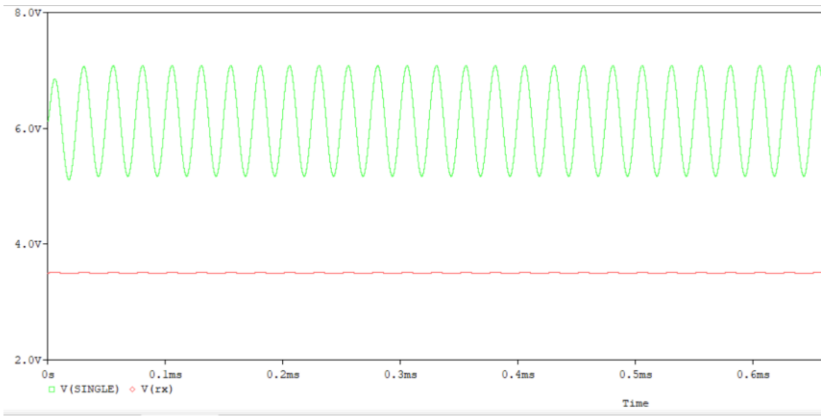


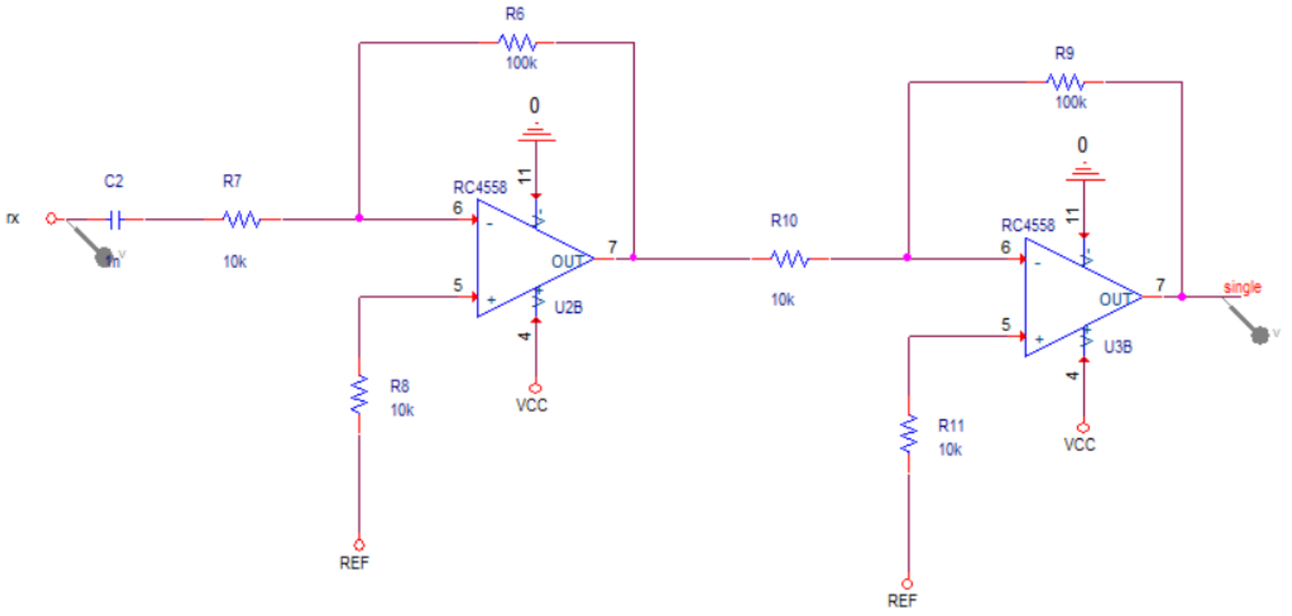


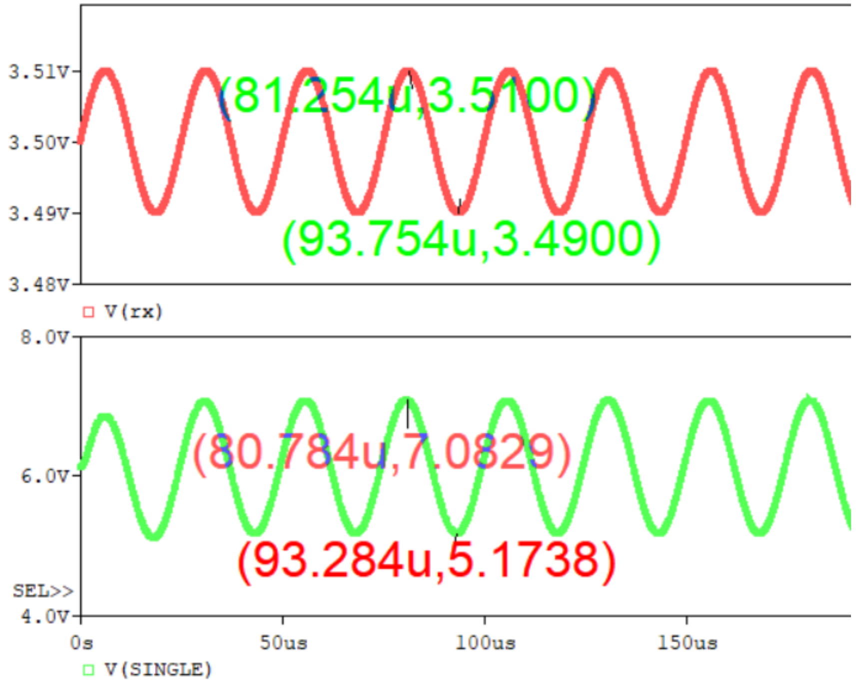
4. Ultrasonic Signal Receiver











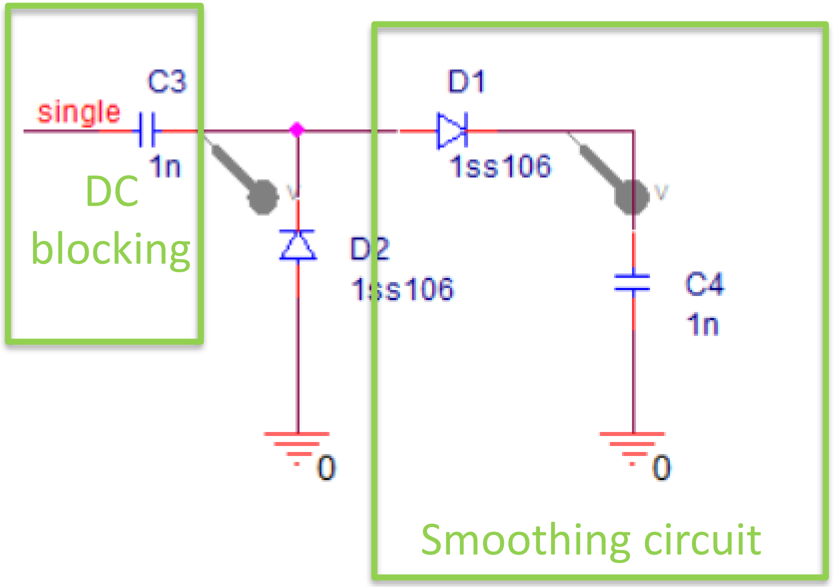
5. Envelope Detector

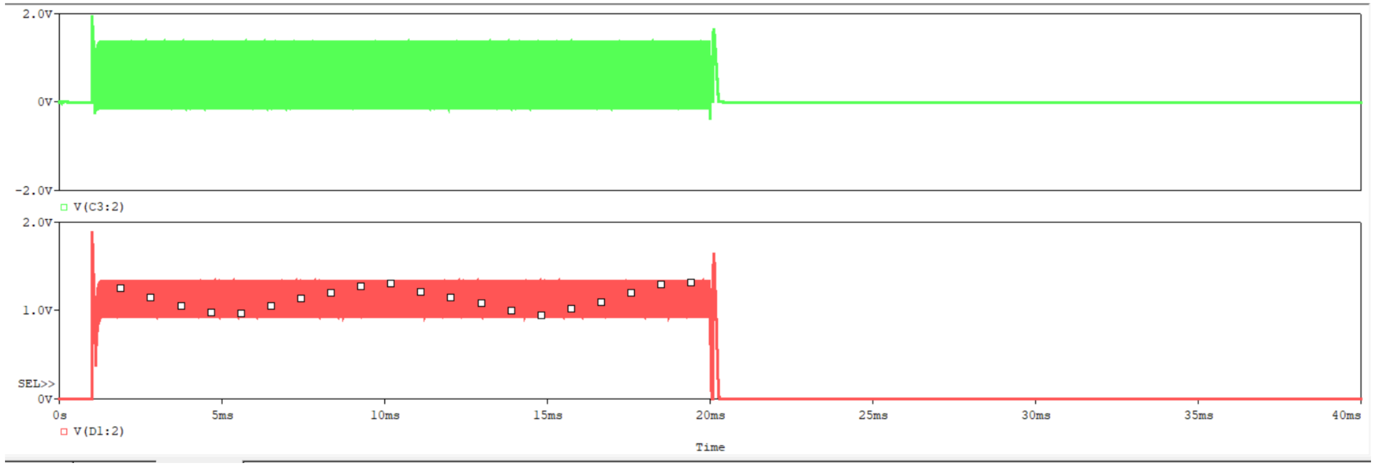
Block Purpose & Function

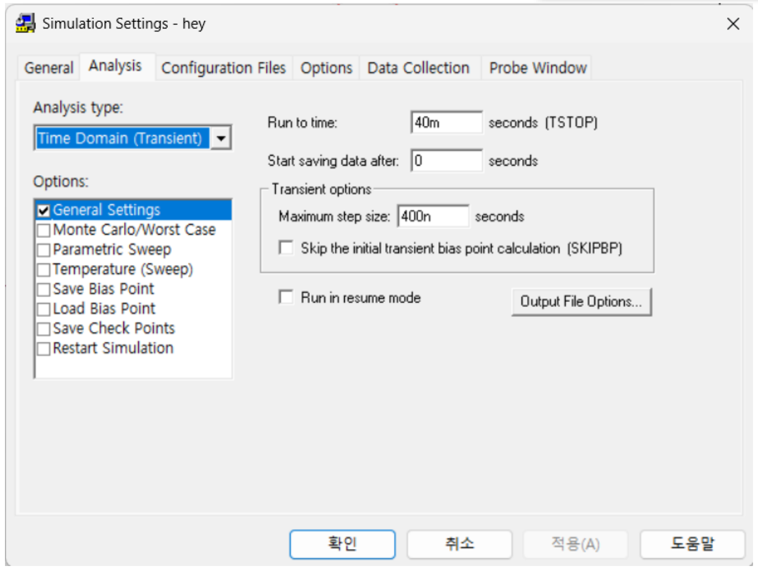
1. Rectify the amplified signal using diodes

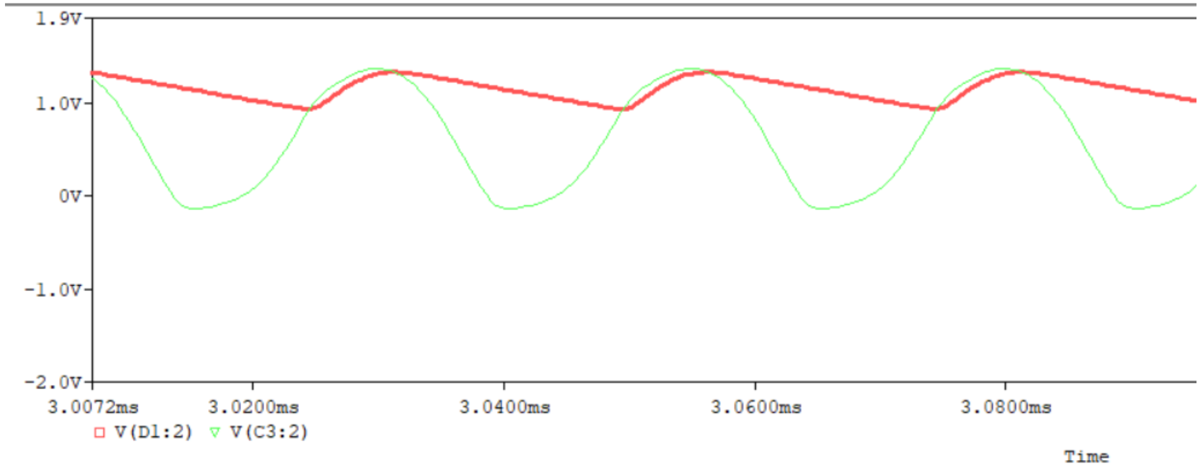
2. Smooth the waveform to extract the envelope (closer to DC)

5. Envelope Detector

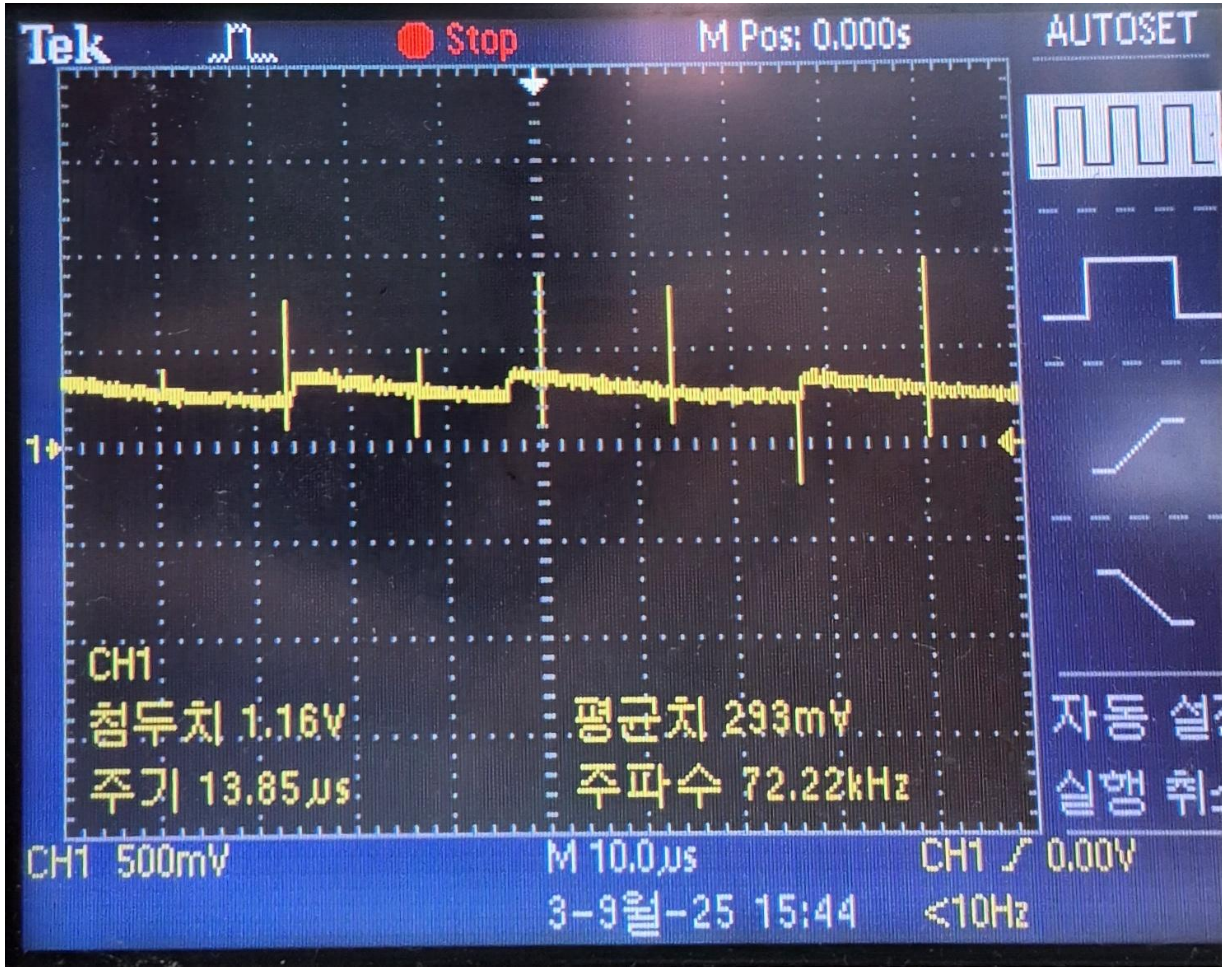


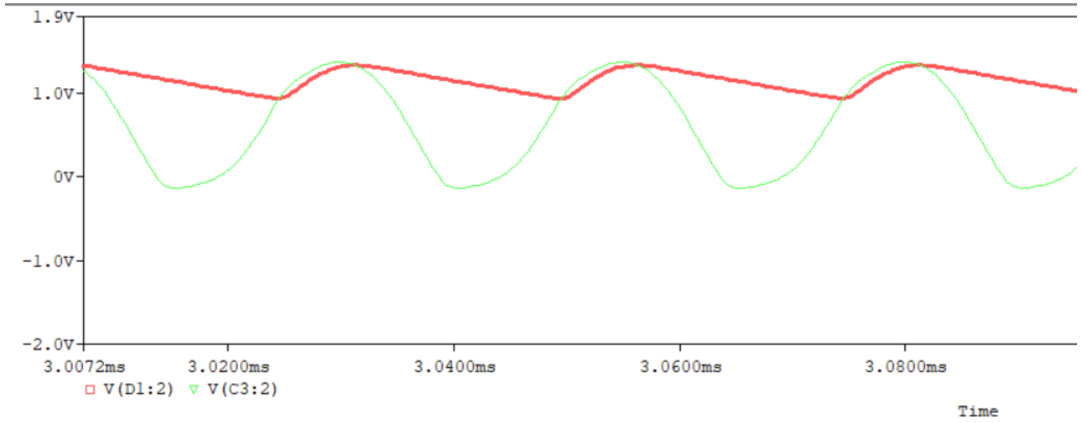






5. Envelope Detector



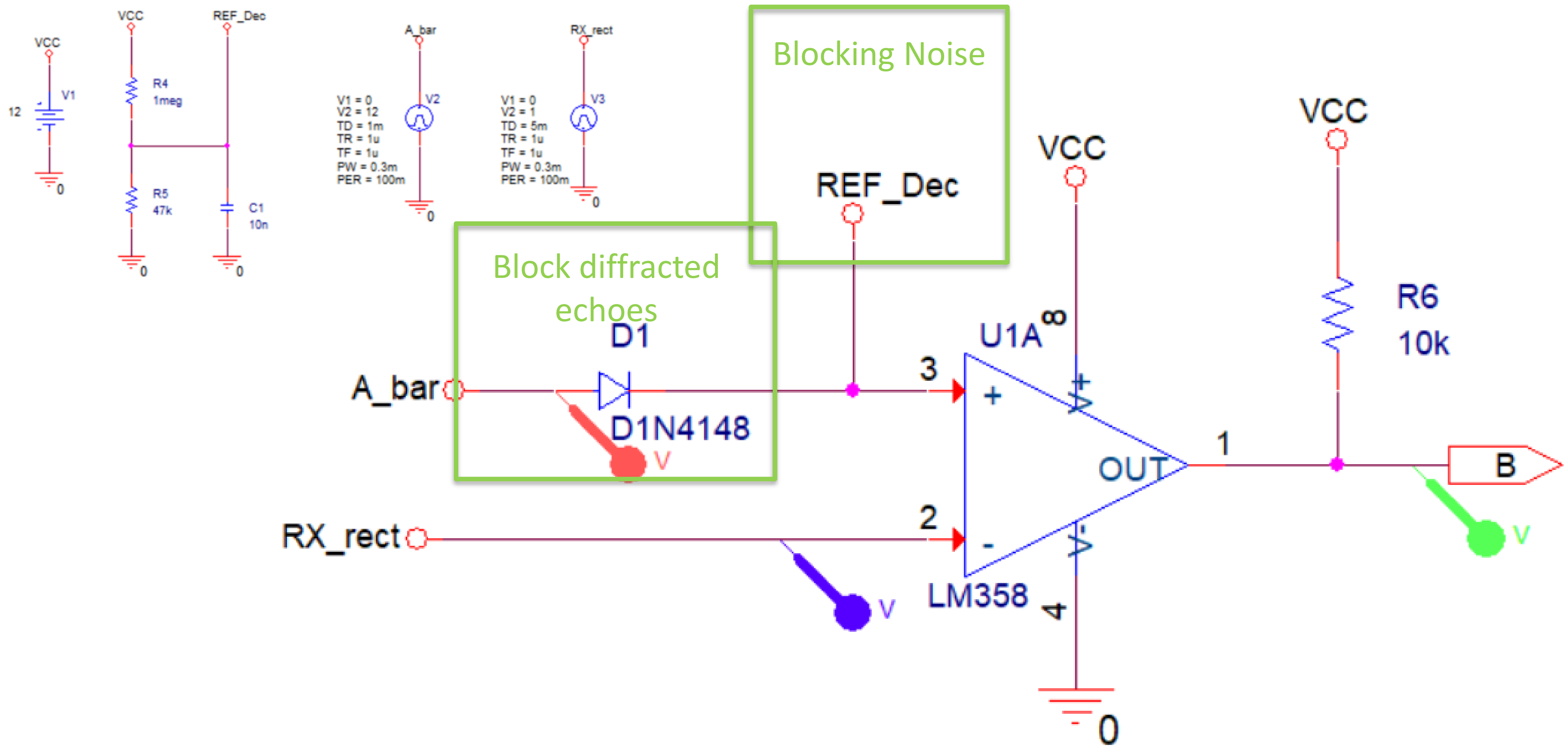


6. Signal Detection (Comparator)

Block Purpose & Function

1. Detect valid received signals using an OP-Amp comparator (active-low output)
2. Block unwanted signals with diode feedback
3. Suppress noise by allowing only signals above ~0.54 V

6. Signal Detection (Comparator)



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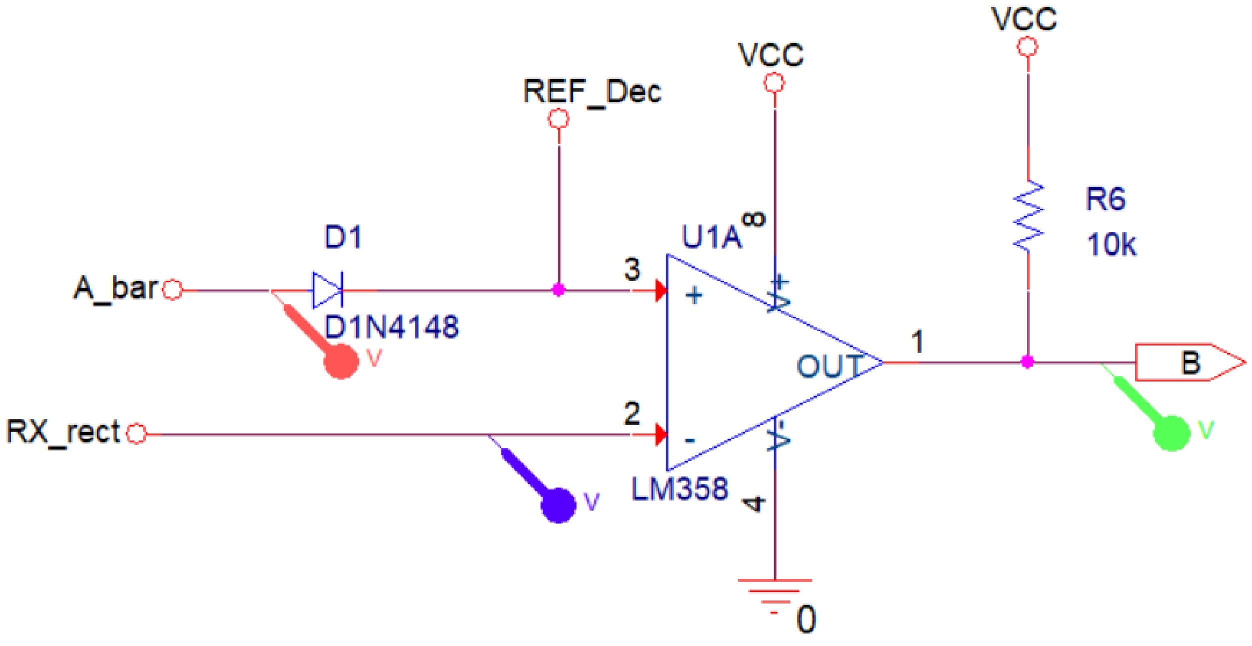
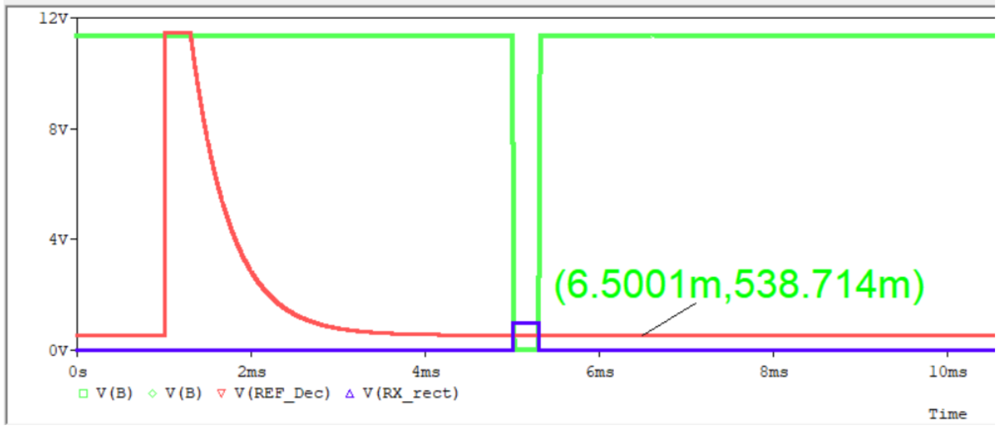




OP-Amp Comparator

•Slow but stable operation

1.V+ > V− → Output = VCC, V+ < V− → Output = GND

2.Signal detection → Output GND (**Active Low**)

Vref

•Noise rejection

→ Detect only signals above ~0.54 V

D1 Diode

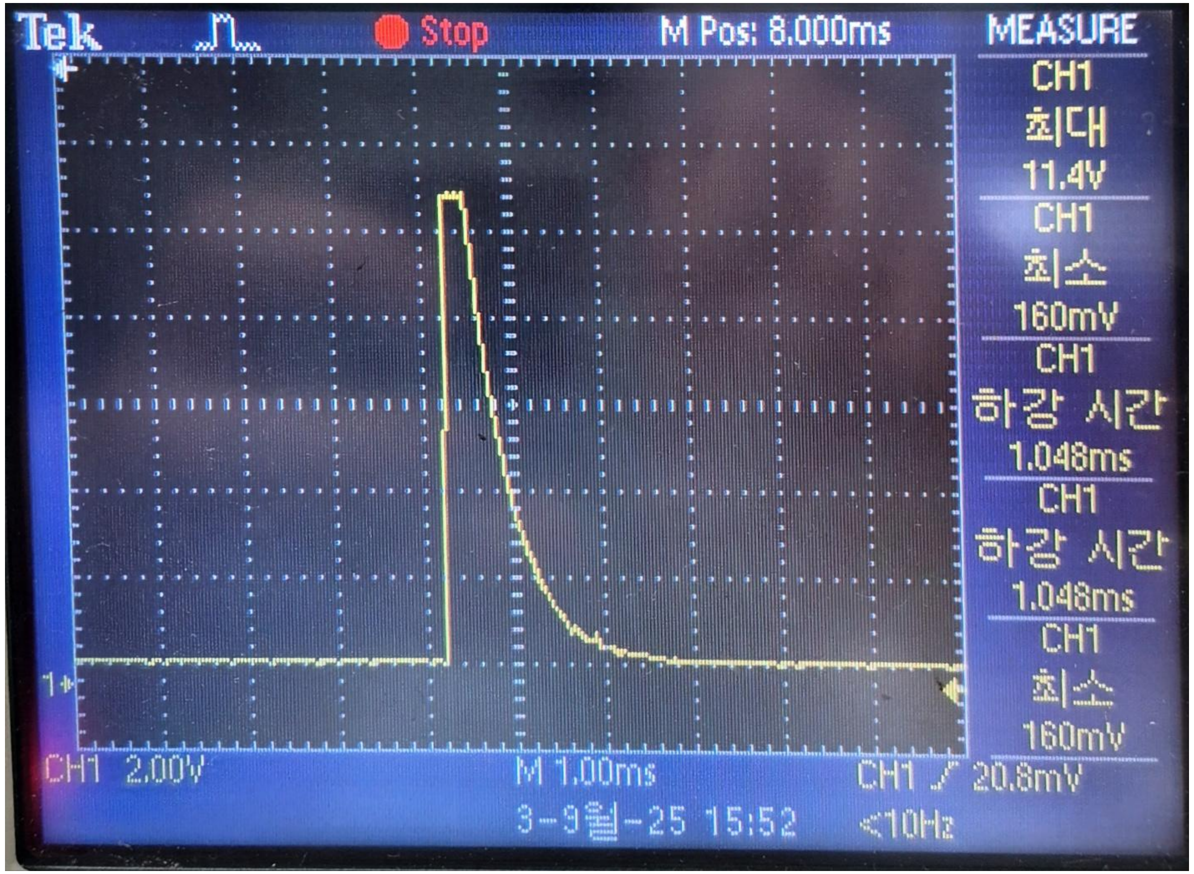
•Blocking diffracted echoes

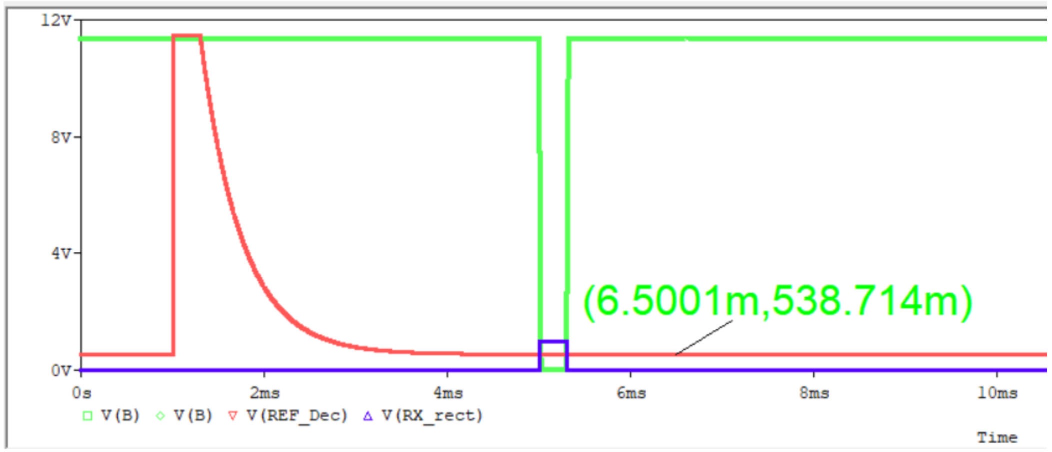
1.Timing pulse → applied to V+ input

2.Capacitor delay → detection time is delayed

3.Ignore signals arriving during this time (diffracted echoes)

6. Signal Detection (Comparator)



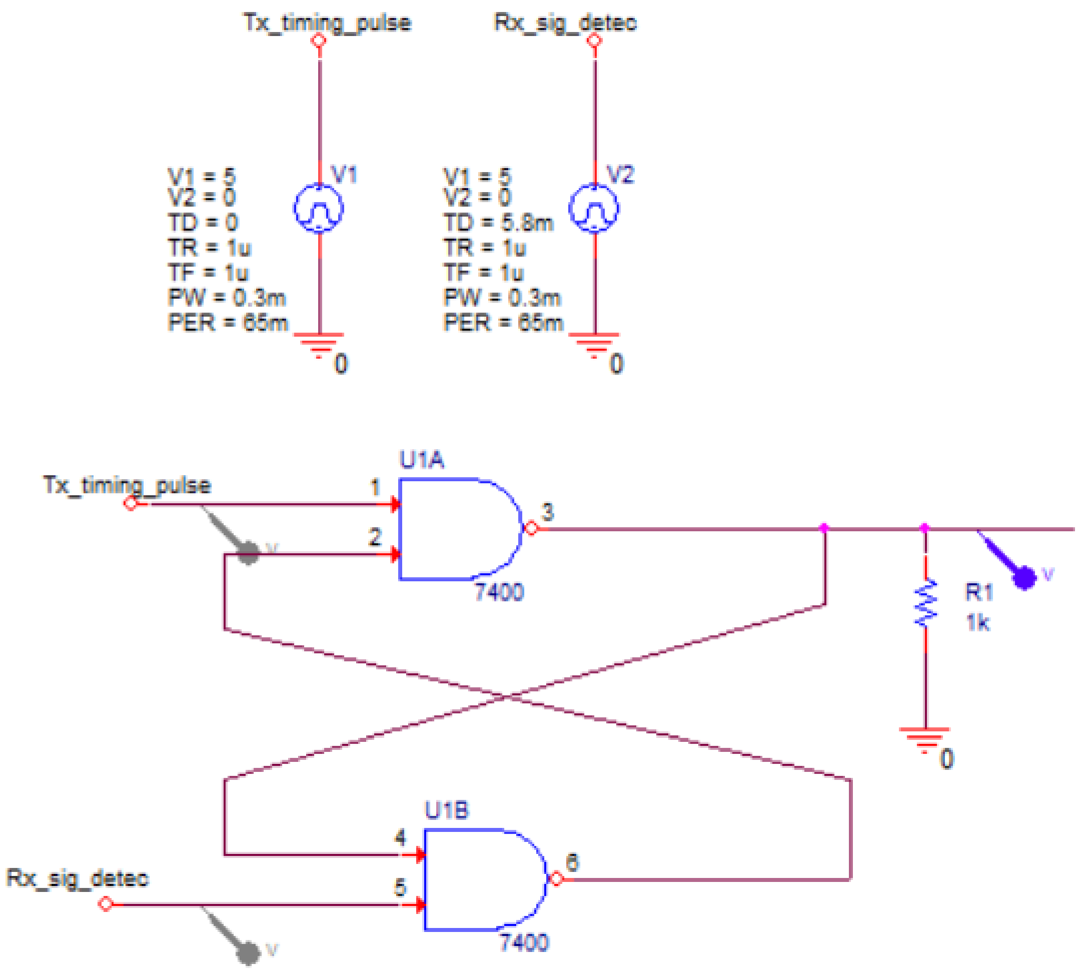


7. Tx/Rx Timing Measurement (SR Latch)

Block Purpose & Function

1. Measure the time difference between transmission and reception
2. Use SR latch to generate a gate signal for the counter

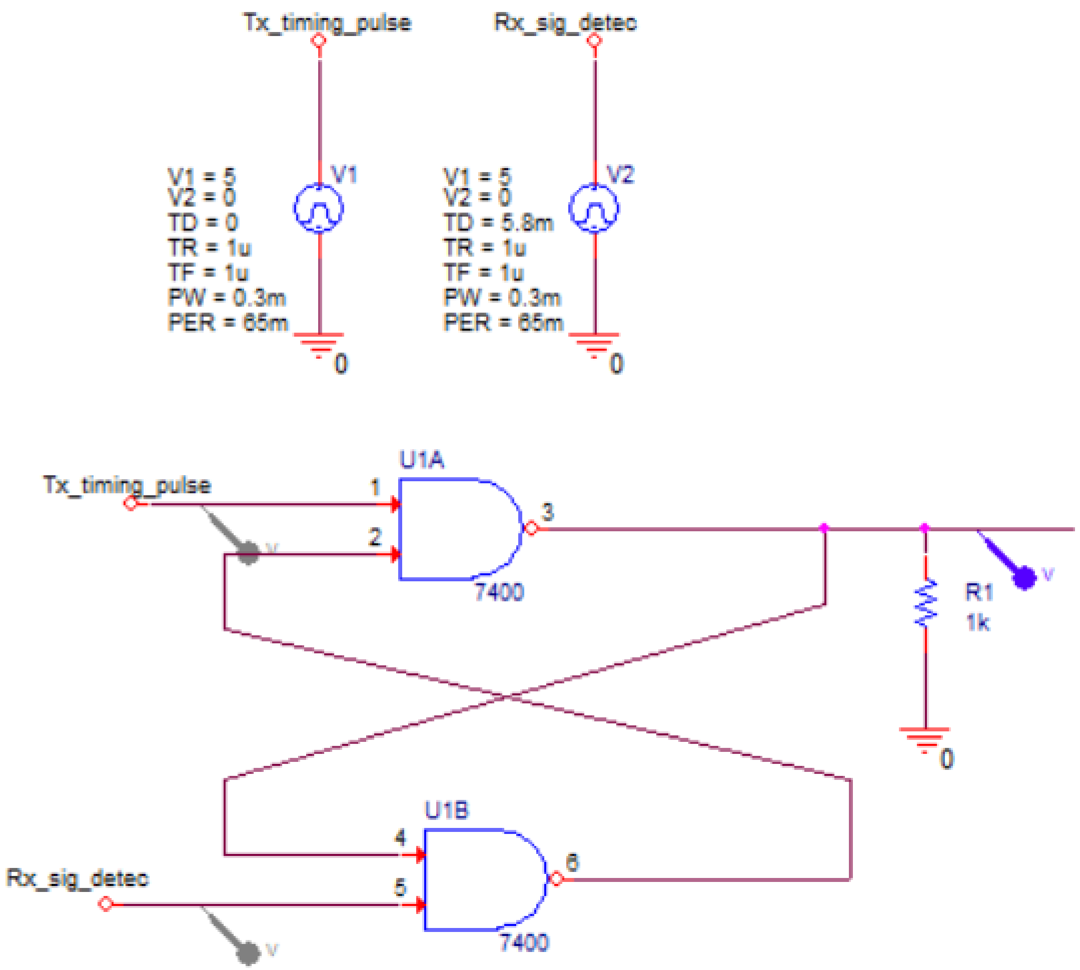
7. Tx/Rx Timing Measurement (SR Latch)

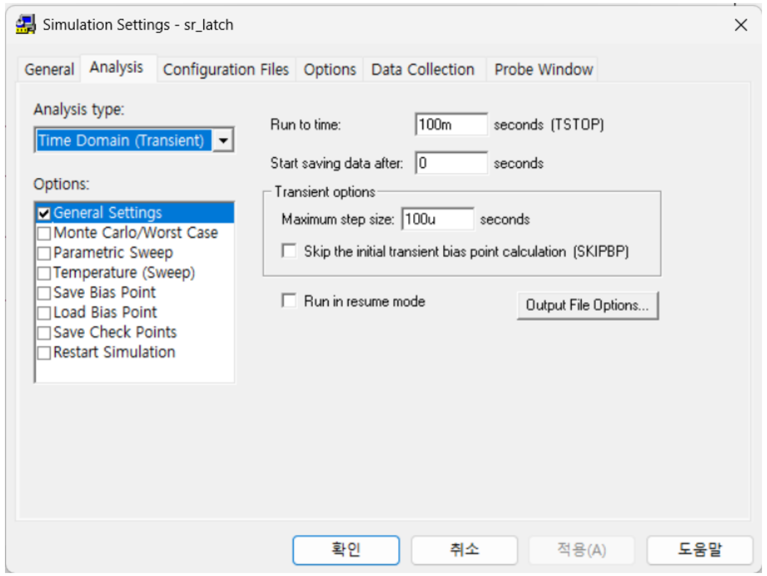


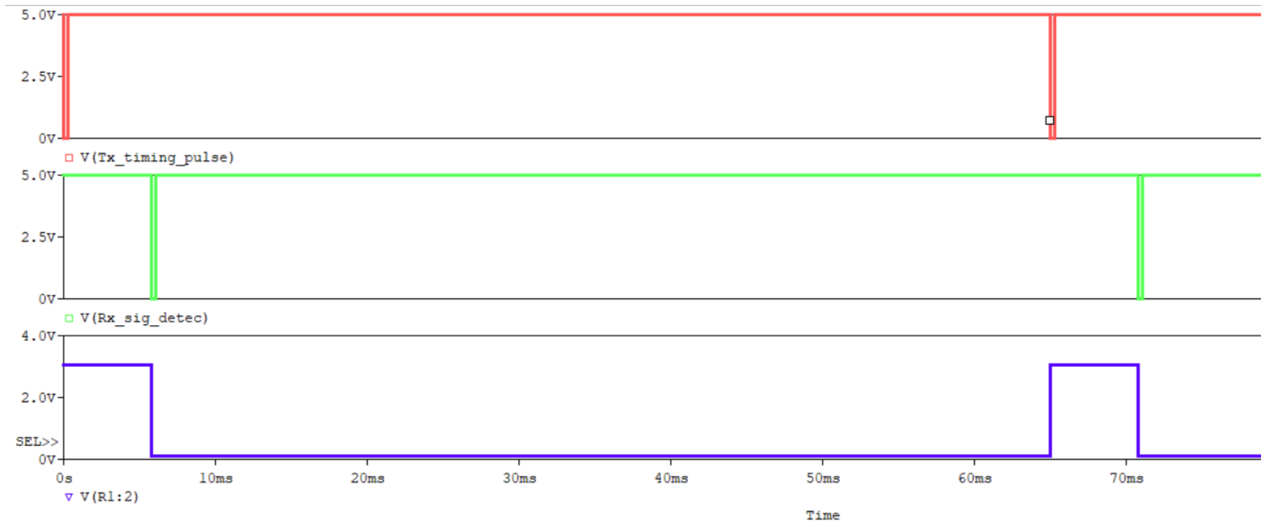
S	R	Q	Q_bar	Operation
1	1	Q	Q_bar	Hold
0	1	1	0	Set
1	0	0	1	Reset
0	0	Invalid	Invalid	Forbidden

<SR Latch Truth Table>

7. Tx/Rx Timing Measurement (SR Latch)







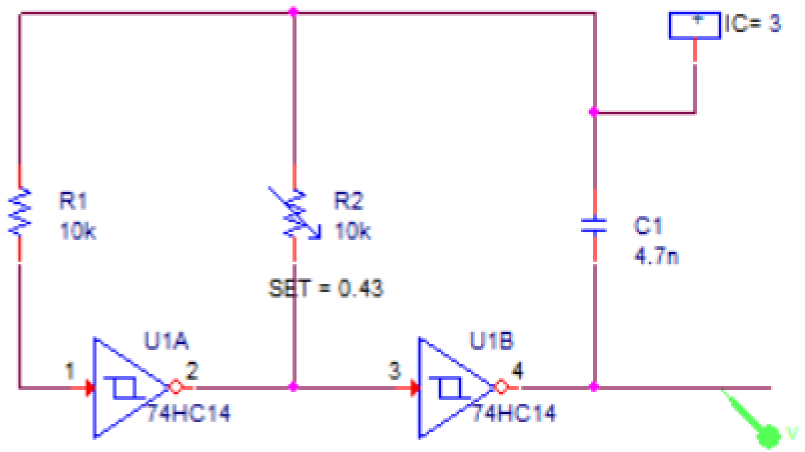
SR Latch High Output → Tx–Rx time interval

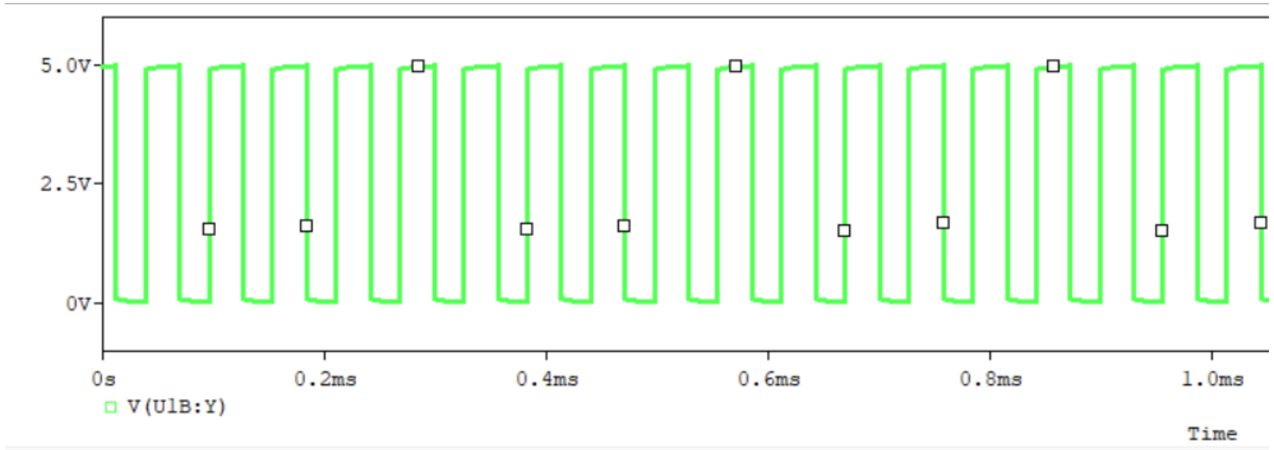
8. Counter Clock Generator

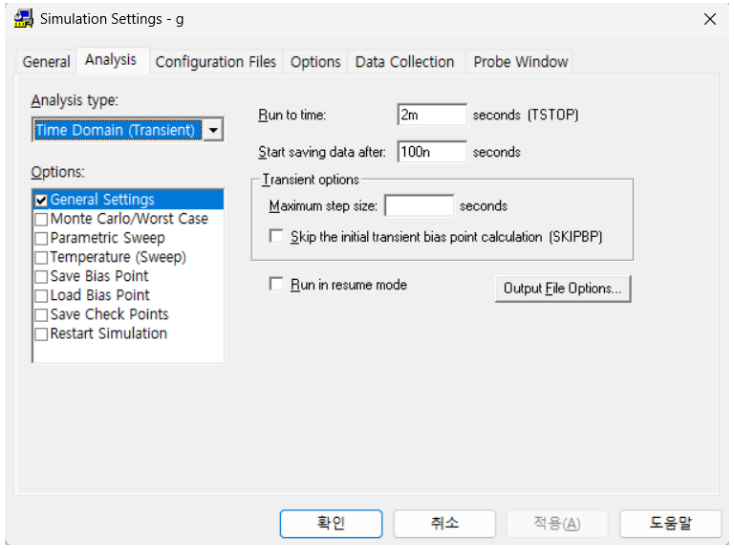
Block Purpose & Function

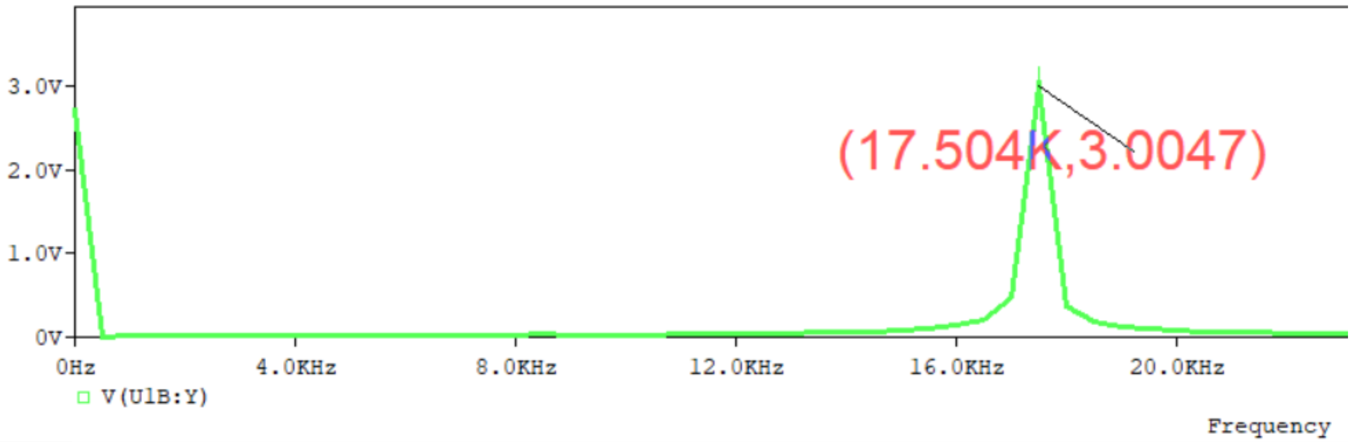
1. Generate 17.18 kHz clock pulses so that 1 m corresponds to 100 counts
2. Supply pulses only during the Tx–Rx timing gate interval

8. Counter Clock Generator

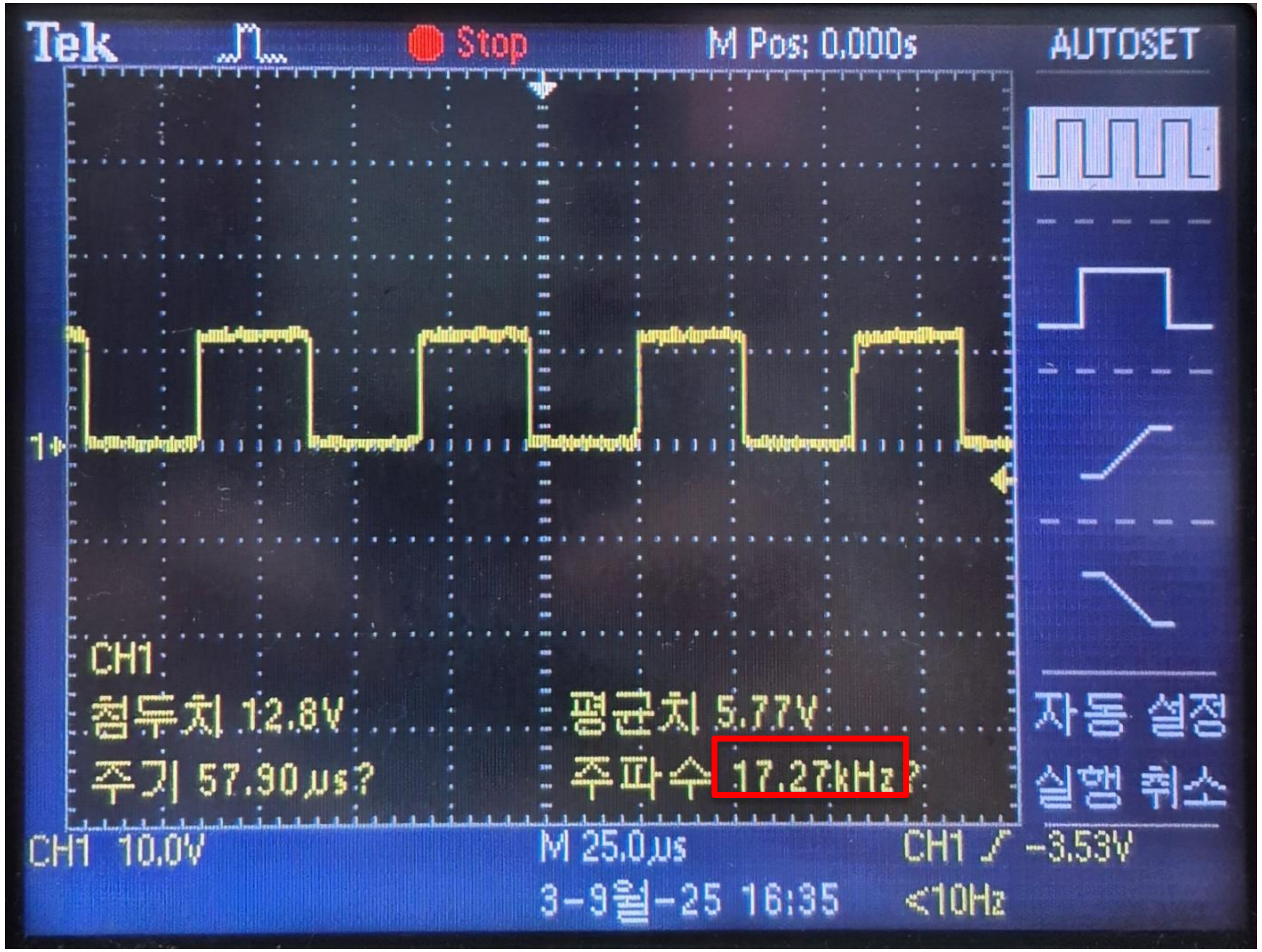


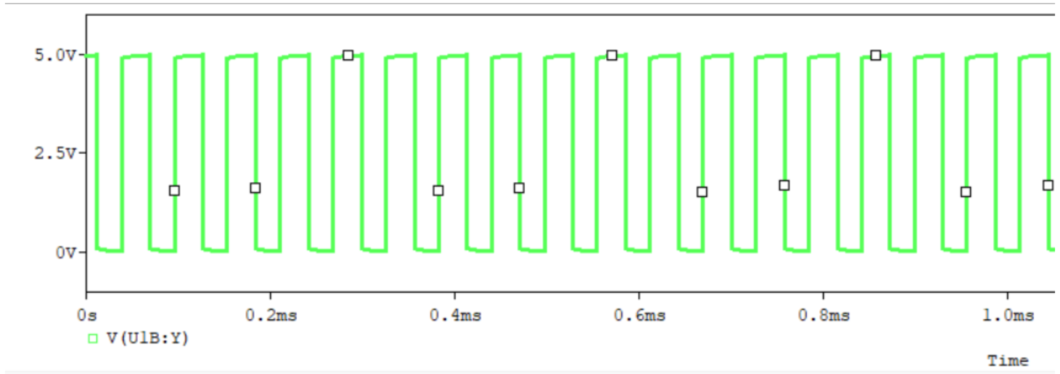


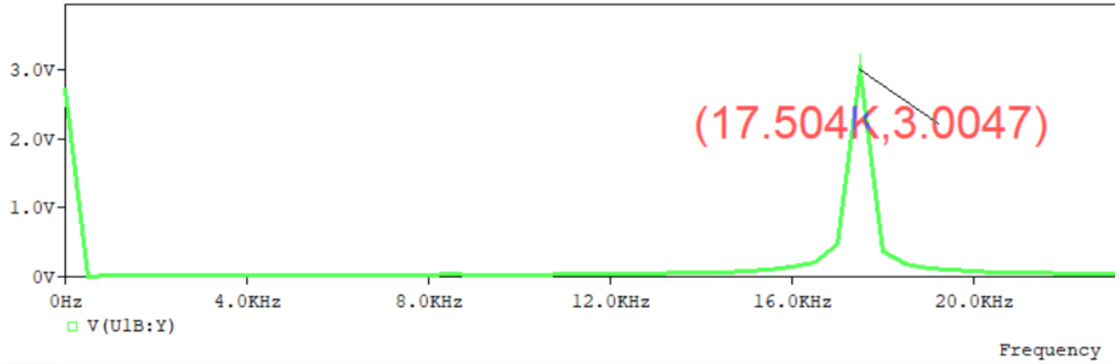




8. Counter Clock Generator





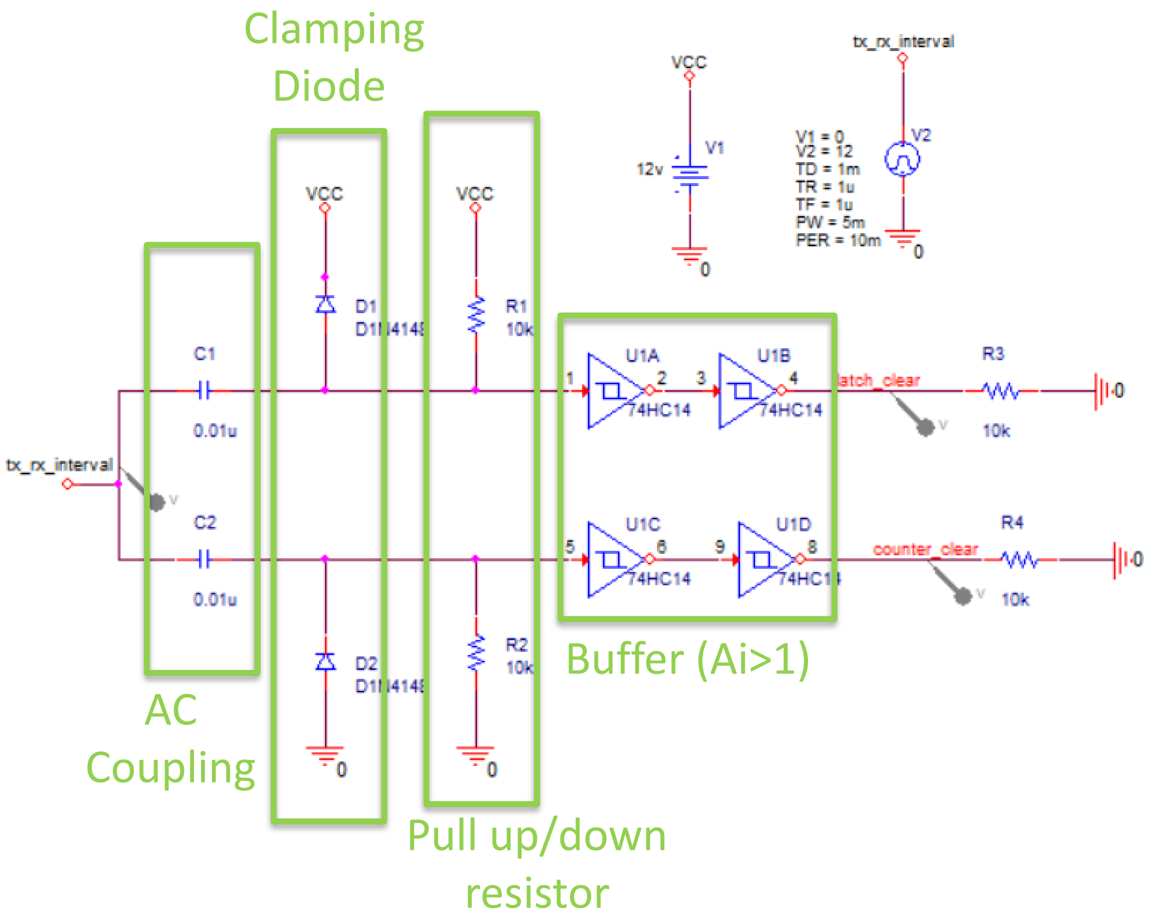


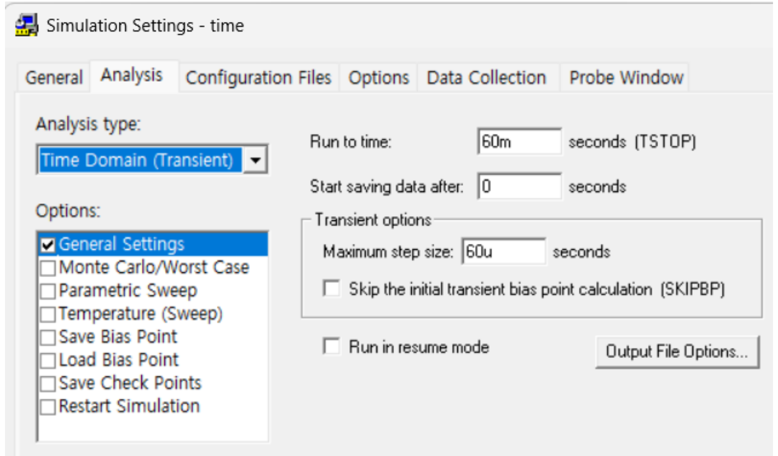
9. Counter Clear & Latch Clear Pulse Generator

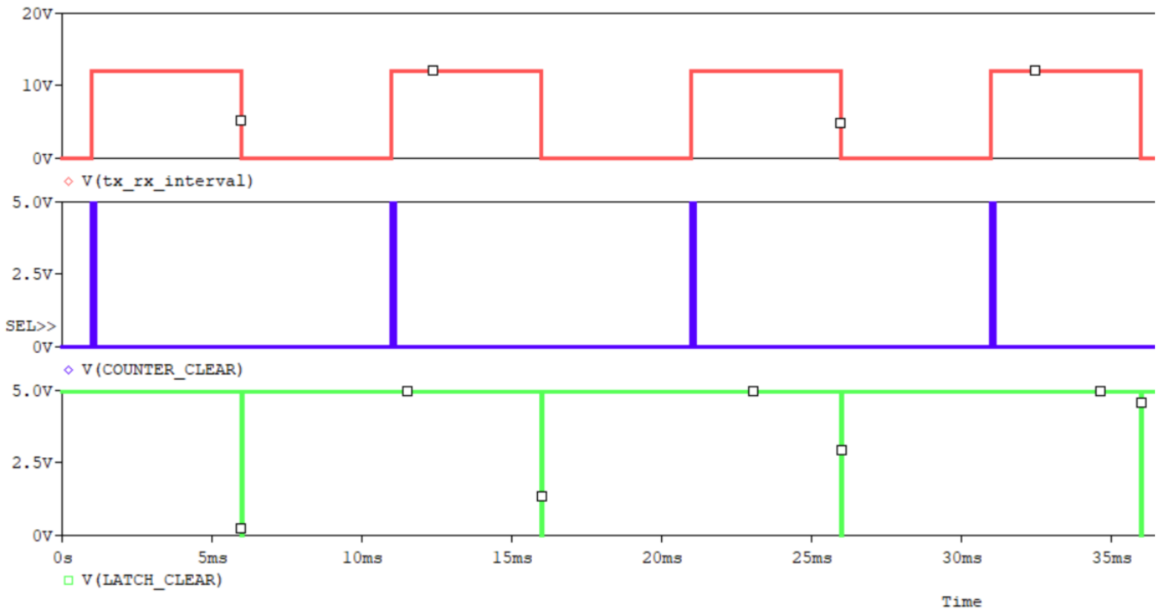
Block Purpose & Function

1. Generate control pulses for counter operation during measurement
2. Provide a counter clear pulse at transmission (Tx) to reset counting
3. Provide a latch clear pulse at reception (Rx) to store the measured value

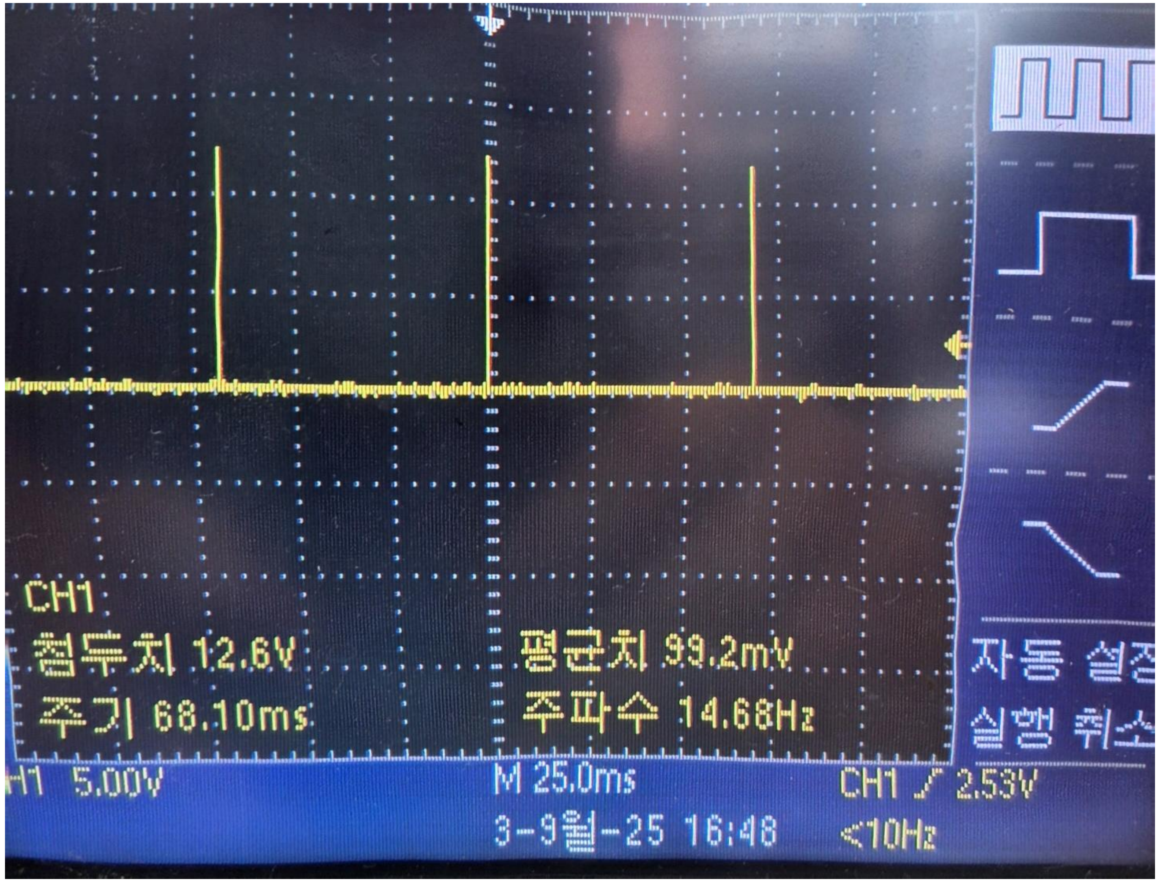
9. Counter Clear & Latch Clear Pulse Generator



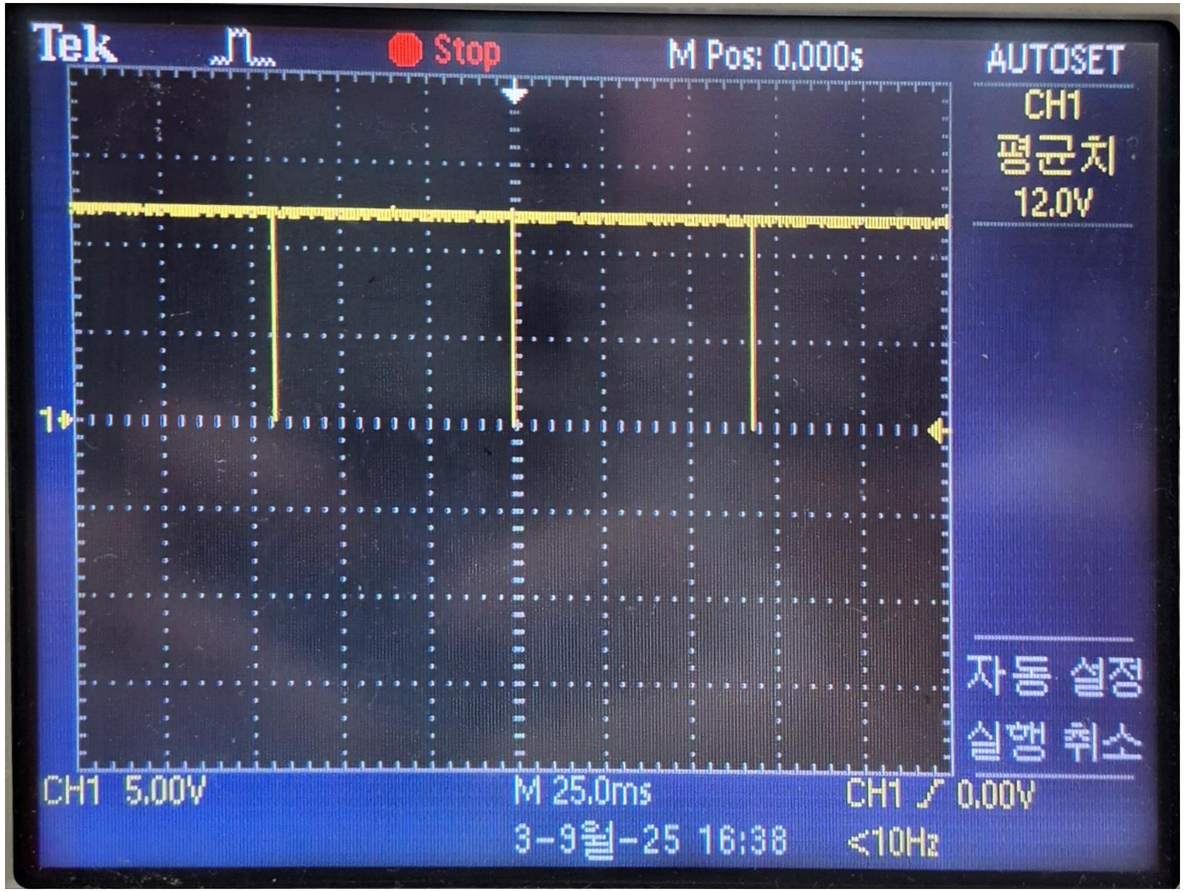




9. Counter Clear & Latch Clear Pulse Generator



<Counter Clear Pulse>



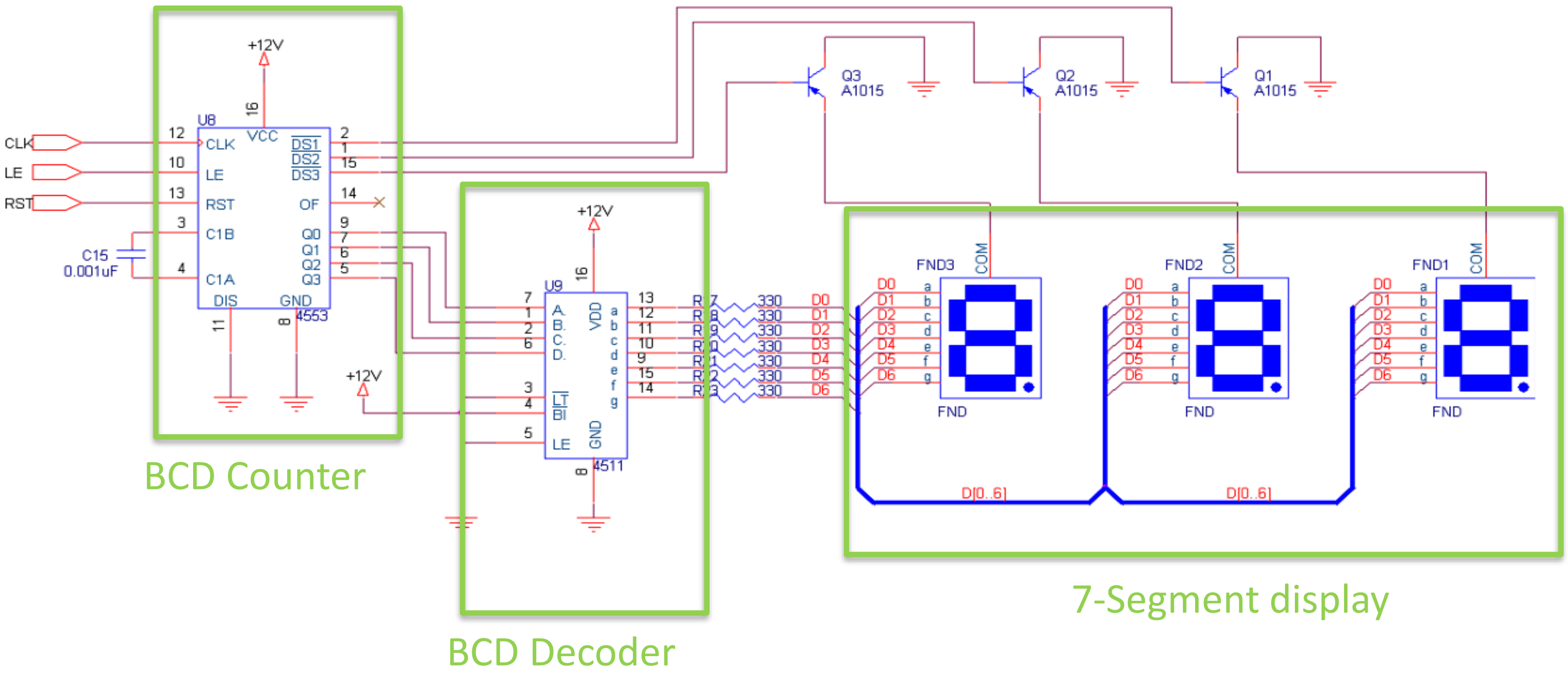
<Latch Clear Pulse>

10. Time Measurement Unit (Counter / Decoder) & 7-Segment Display Output (FND)

Block Purpose & Function

1. Count pulses during the Tx–Rx timing interval to calculate distance
2. Decode the count and display the measured distance on 7-Segment FND

10. Time Measurement Unit (Counter / Decoder) & 7-Segment Display Output (FND)

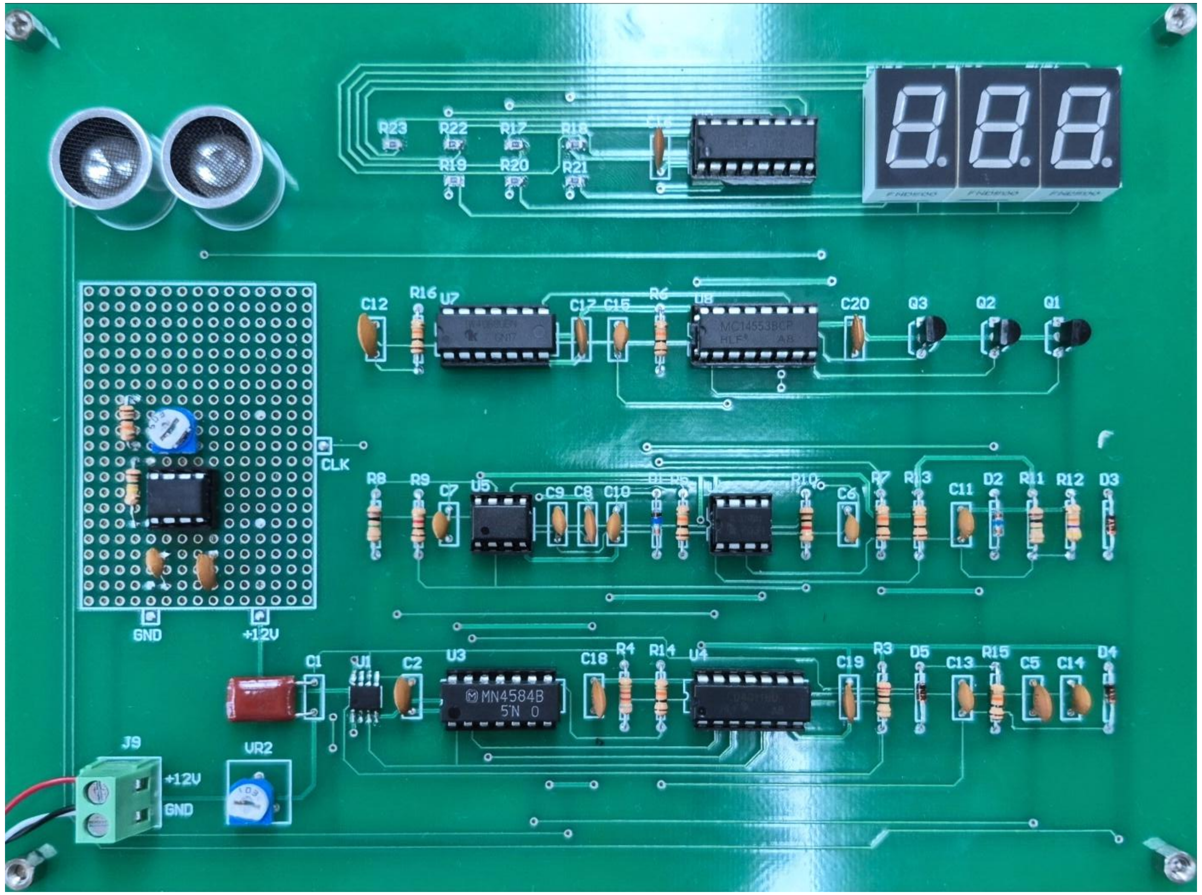


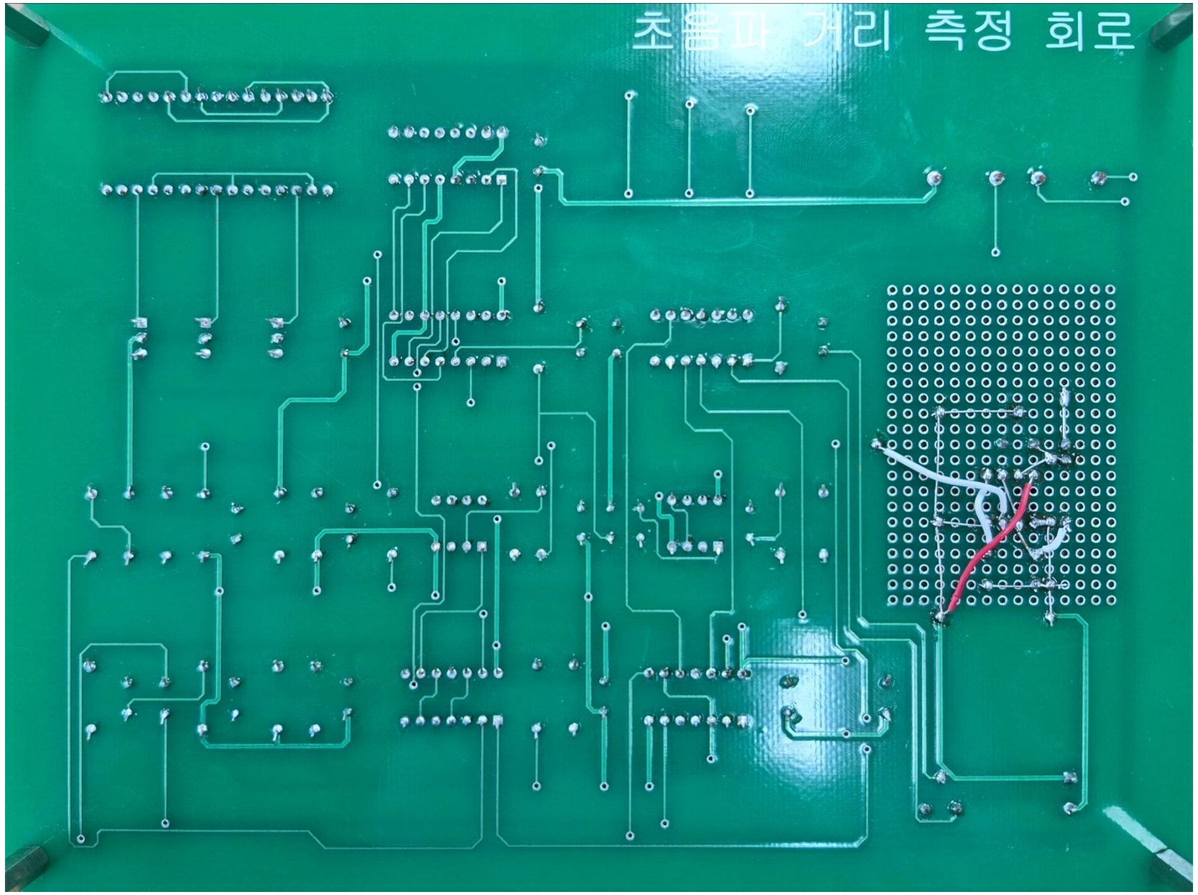
Count pulses during the Tx–Rx interval using a BCD counter.

Use a BCD decoder to convert the count into segment signals.

Drive the 7-Segment display (FND) to show the measured distance.

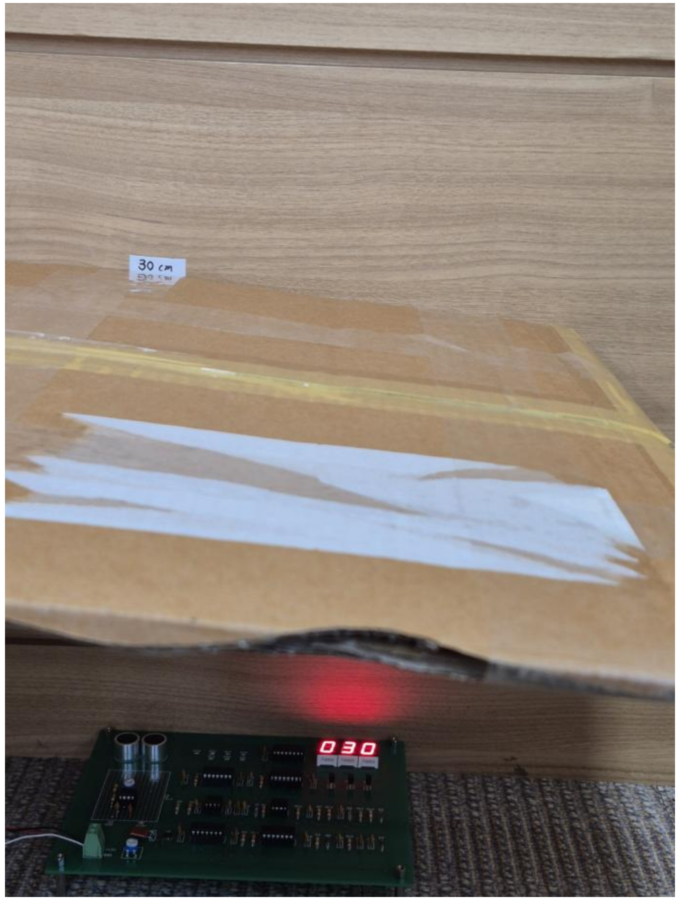
PCB Circuit Photo





PCB & Demo photos

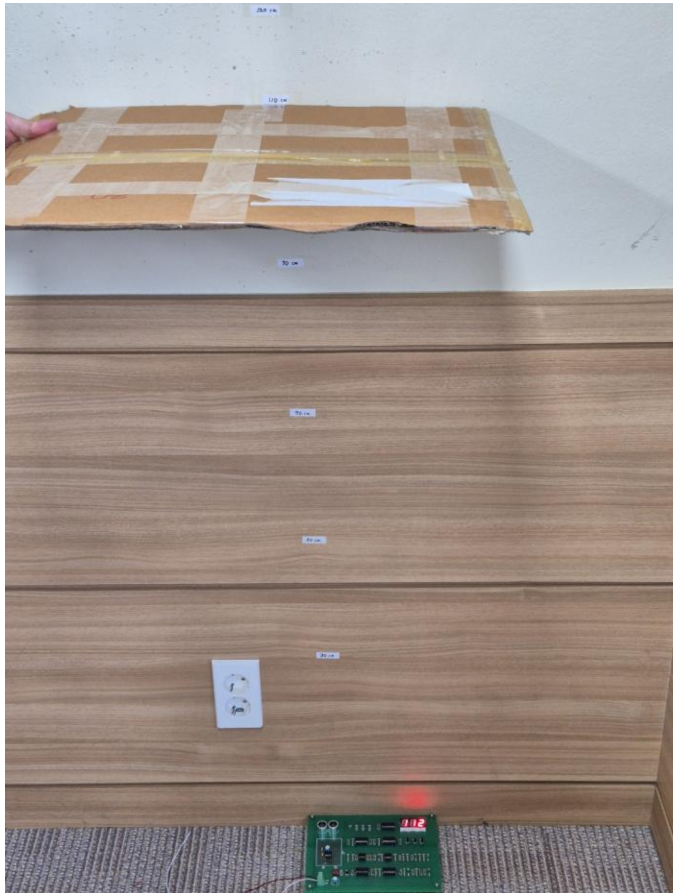
Demo Photos



Distance	30	50	70
Measurement result	30	51	70
Error	0	1	0

PCB & Demo photos

Demo Photos



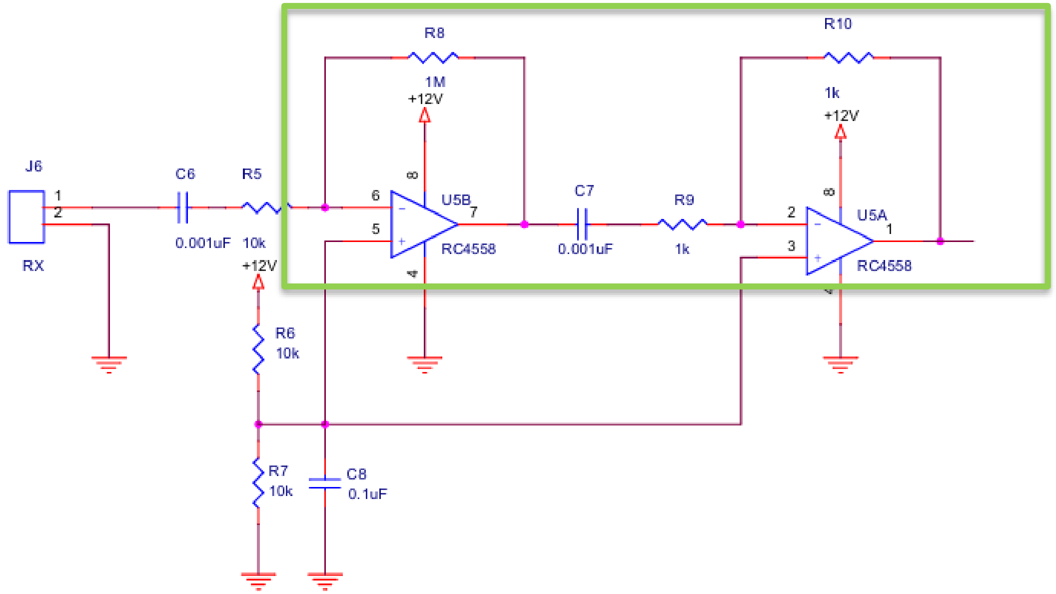
Distance	90	110
Measurement result	90	112
Error	0	2

Issues & Improvements

1) OP-amp gain shortfall
2) No ESD protection on the supply I/O
3) Grounding problems (mixed/unstable GND)
4) Metastability (async control vs. clock)
5) No resistors on FND drivers
6) NE555 RC Oscillator — Temperature Drift
7) Low flexibility/expandability

Issues & Improvements

1. OP-amp gain shortfall

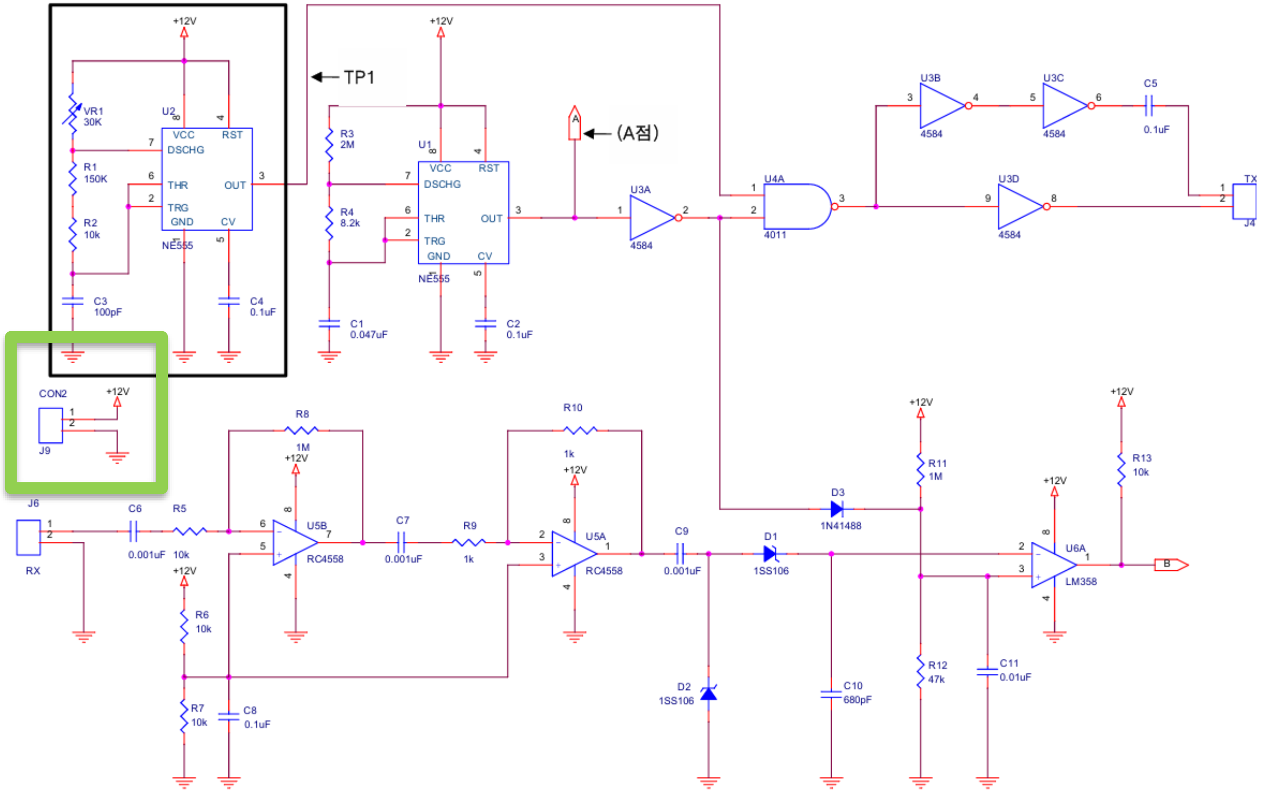


Issue :
Gain = GBW / BW = 3 MHz / 40 kHz ≈ 75 → 100× gain is not achievable (in one stage).

Improvement :
Reconfigure to [×10] + [×10] two-stage op-amp → overall ×100 gain achievable (stable).

Issues & Improvements

2. No ESD protection on the supply I/O



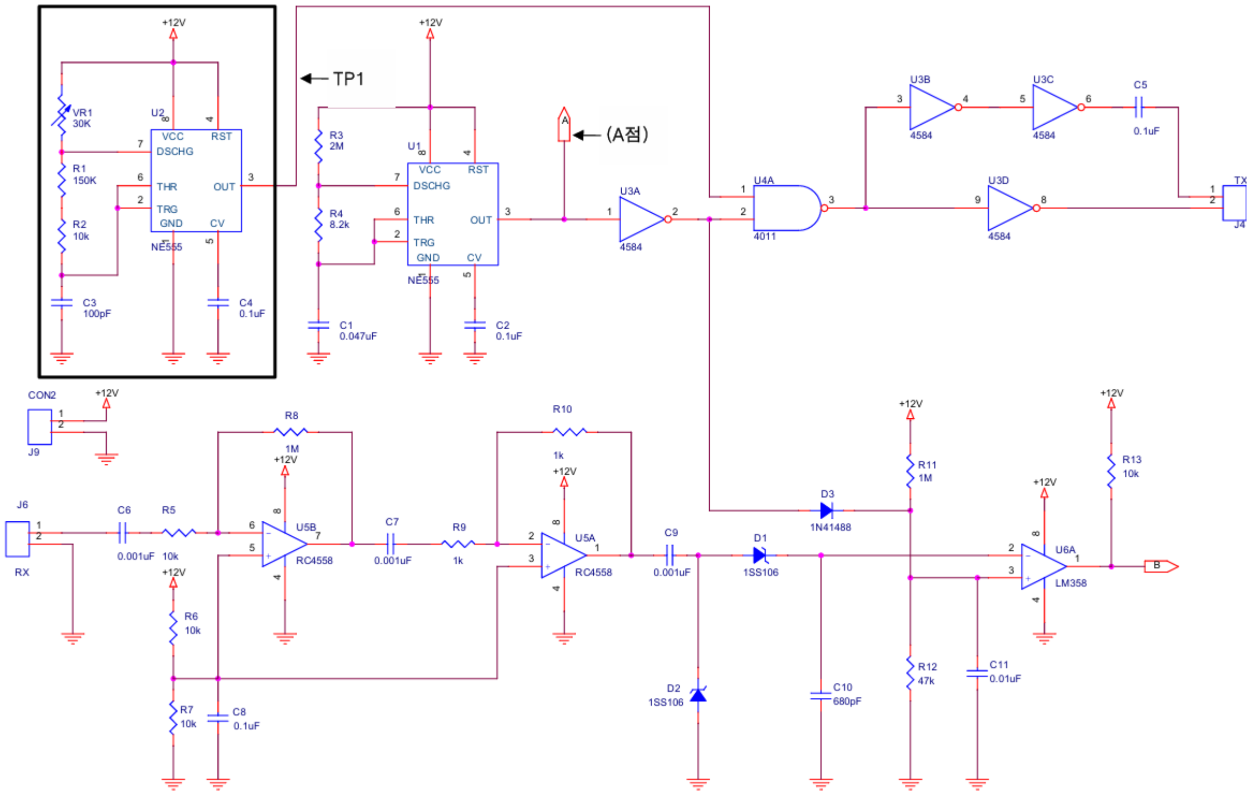
Issue:

- **Surges/ESD** can stress or damage the circuit.

Improvement :

- Add **TVS** diodes at supply inputs and external connectors (series R / ferrite as needed).

3. Grounding problems (mixed/unstable GND)



Issue:

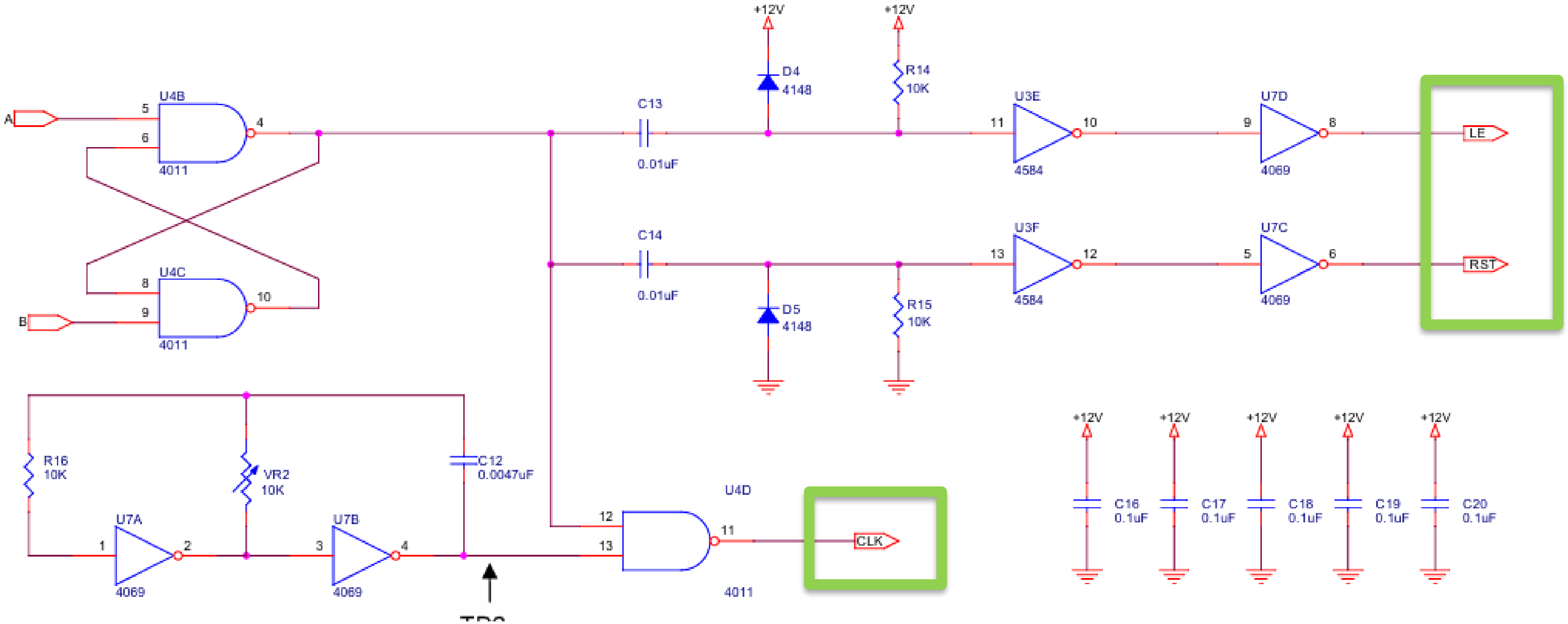
- Power and signal grounds are not separated → noise coupling, unstable reference. EMI increase

Improvements:

- **Split power vs. signal GND**, use **ferrite beads + decoupling caps**, and a **copper ground pour** .

Issues & Improvements

4. Metastability Problem (async control vs. clock)



Issue:

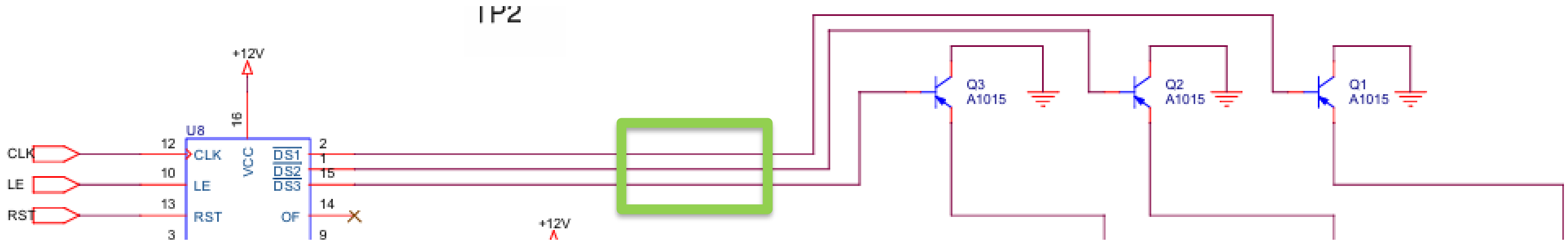
- Control signals and the counter clock are not synchronized → wrong counts.

Improvements:

- **Synchronize with 2-FF synchronizers** and use a synchronous clock
- option: **172 kHz clock** for 0.1-cm resolution,
- Use an **MCU** to synchronize the gate/clock and average multiple readings (output the mode).

Issues & Improvements

5. No resistors on FND drivers



Issue:

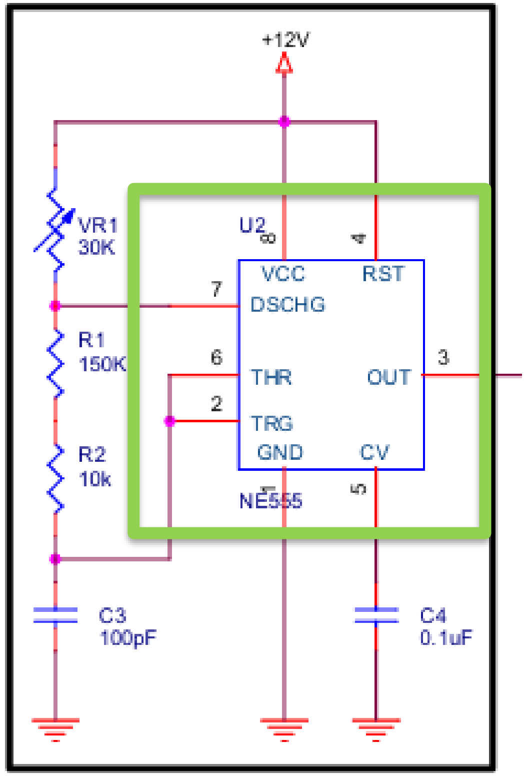
- Missing base series and pull resistors → stress, reduced lifetime/ghosting.

:

- Add **series base resistors** and proper pull-up/pull-down; limit segment current.

Issues & Improvements

6. NE555 RC Oscillator — Temperature Drift



Issue:

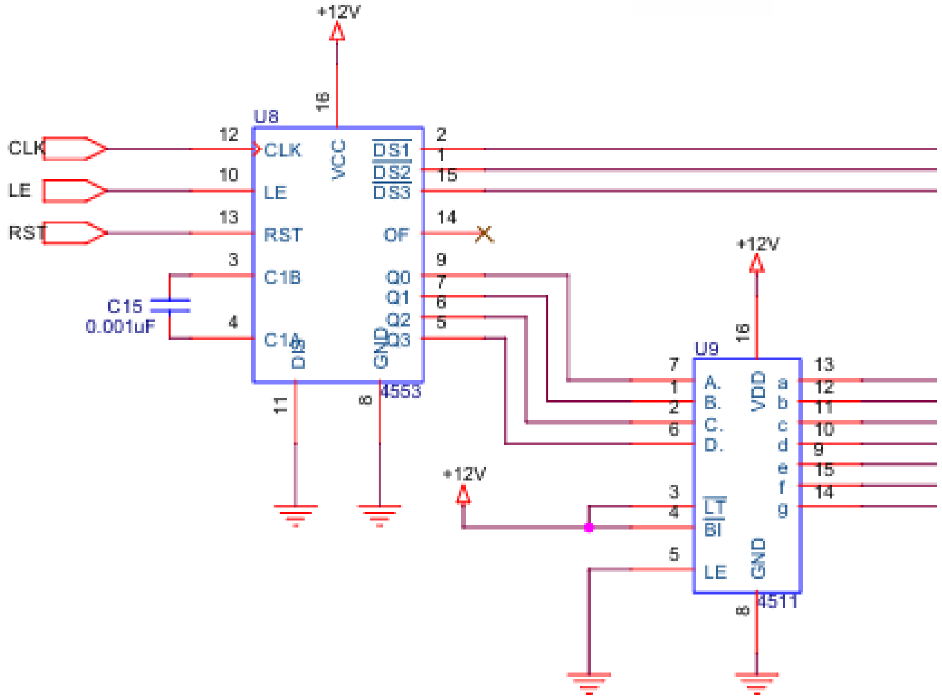
- The NE555-based RC oscillator shows significant frequency shift with temperature (40 kHz varies).

Improvement:

- Use a **crystal oscillator** (or TCXO) for stable frequency.

Issues & Improvements

7. Low flexibility/expandability



Issue:

- Hard to change features with discrete logic only.

Improvements:

- Use a low-cost **MCU** to integrate timing, gating, counting, filtering, and display; firmware calibration and synchronized clocks.

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